

An introduction to Cavity Quantum Materials

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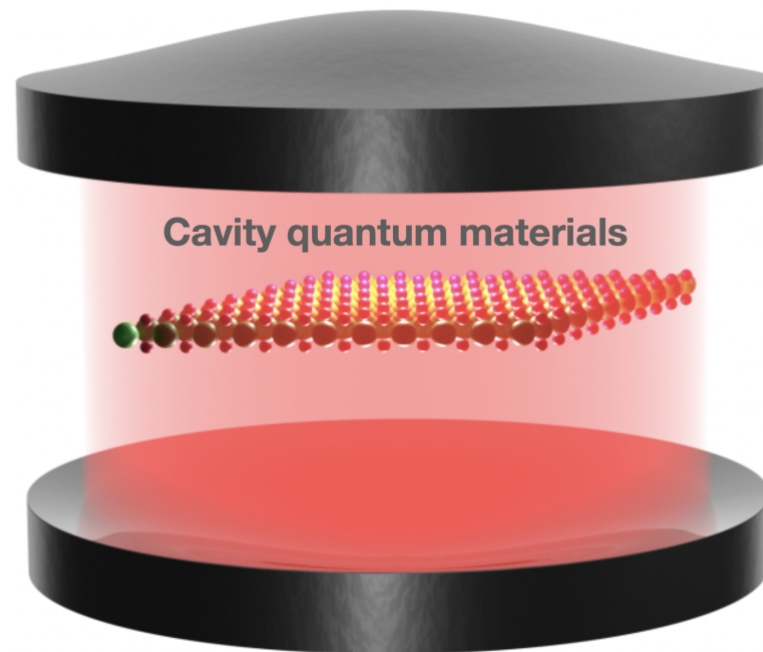


Outline

- **Motivation:** What do we mean by CQM? Where do we aim and why cavities?
- **Floquet engineering:** what it is? what did we learn? Hamiltonian design, topology
- **The quantum limit:** what is new? The quantum vacuum fluctuations
- **The quantum Rabi and the Jaynes-Cumming models:** differences? Relevance? Vacuum Rabi oscillations.
- **Solid state systems in a cavity:** the Peierls substitution, two simple examples

Disclaimer: *This is a lecture biased by the taste of the lecturer, not at all a state of the art presentation of the subject... but, hopefully, it will motive you to dip into it*

What is a CAVITY QUANTUM MATERIAL?



In short: a solid state system inside a cavity in the quantum regime

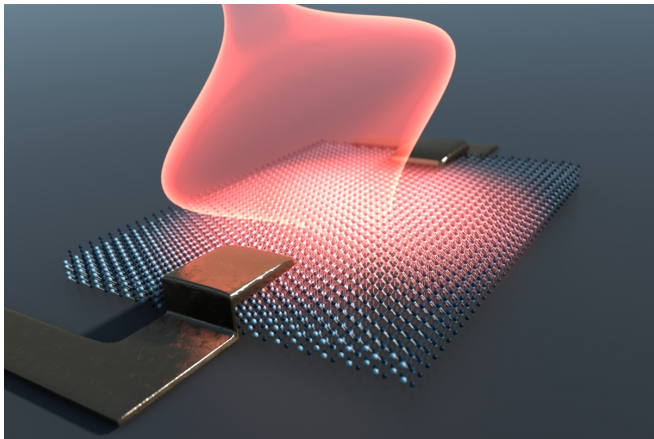
- quantum cavity (very few quanta)
- better name: quantum cavity materials..(?)

Usually materials with an energy dispersion (not isolated atoms.. not a collection of TLS)
Cavity not necessarily photons (also phonons or magnons... or any boson)

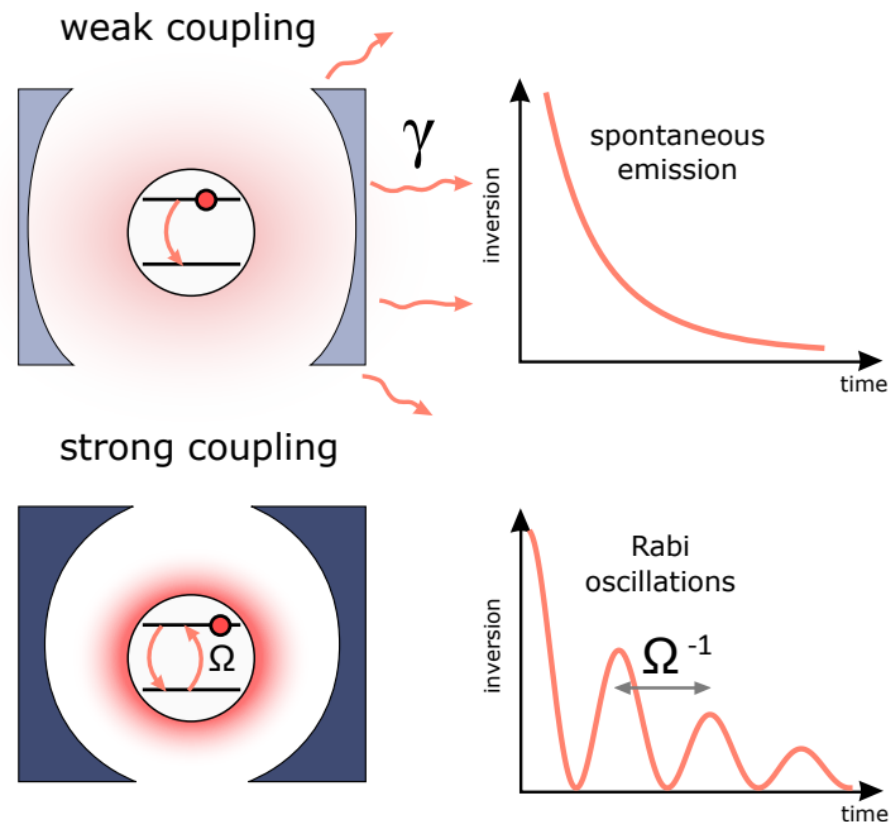
Motivation

Our (general) goals:

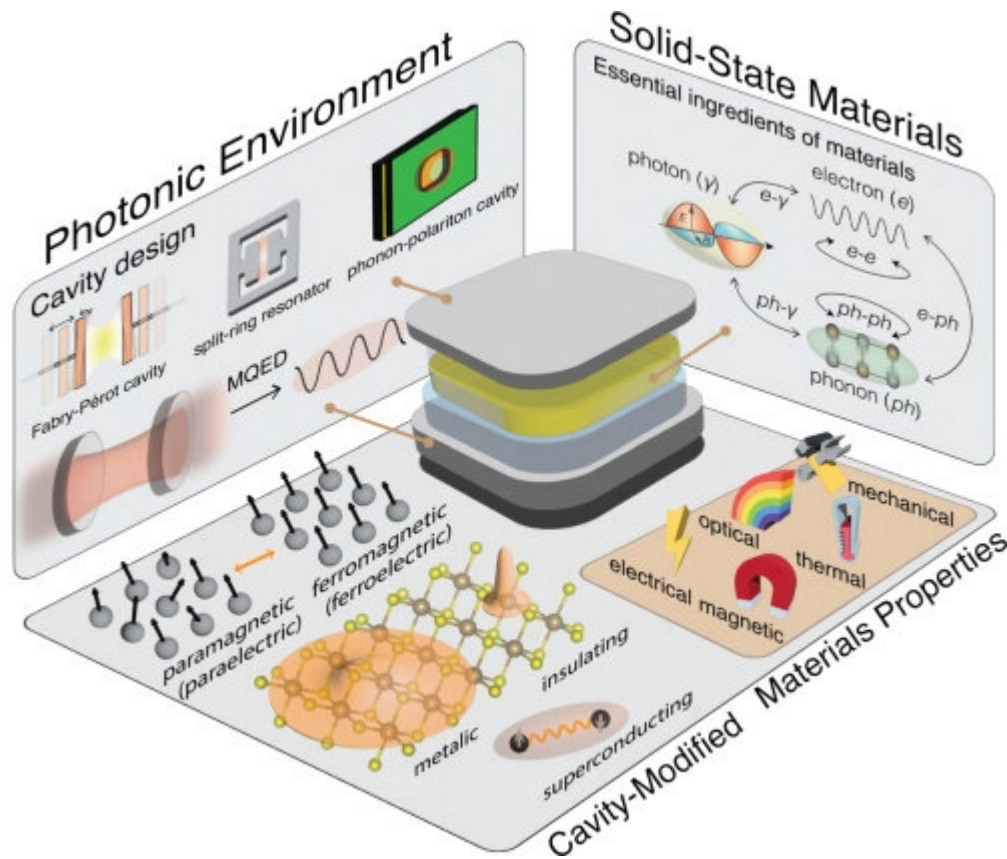
- understand nature,
- find new phenomena,
- control different properties,
- to find new functionalities,
- ultimate quantum control (QT)



Floquet engineering
(Hamiltonian design, topology)



Cavity quantum materials: what for?



- **Use quantized cavity electromagnetic fields as a control knob for many-body matter.**
Instead of driving materials with classical light, the goal is to change intrinsic material properties (band structure, interactions, order parameters, pairing) by placing the material inside a cavity so the vacuum or a few quantized modes hybridize with electronic/collective degrees of freedom.
- **Engineer or induce new effective interactions and phases.**
Photon-mediated interactions can be attractive or long-range and can, in principle, stabilize new correlated phases (enhanced excitons, photon-mediated pairing, modified magnetism, topological responses). Demonstrating robust, cavity-enabled ground-state or steady-state changes is a central aim.

Quantum limit of the Floquet physics → topology.

A reality?

This is a field mainly driven by theorists and by hard working (and clever) experimentalists

Breakdown of topological protection by cavity vacuum fields in the integer quantum Hall effect

Felice Appugliese^{1*}, Josefine Enkner¹, Gian Lorenzo Paravicini-Bagliani^{1†}, Mattias Beck¹, Christian Reichl², Werner Wegscheider², Giacomo Scalari¹, Cristiano Ciuti³, Jérôme Faist^{1*}

PHYSICAL REVIEW LETTERS **131**, 196602 (2023)

Weakened Topological Protection of the Quantum Hall Effect in a Cavity

Vasil Rokaj^{1,2,*}, Jie Wang^{2,3,†}, John Sous^{4,5,6}, Markus Penz⁷, Michael Ruggenthaler⁸ and Angel Rubio^{8,9,‡}

Vacuum Bloch–Siegert shift in Landau polaritons with ultra-high cooperativity

Xinwei Li, Motoaki Bamba, Qi Zhang, Saeed Fallahi, Geoff C. Gardner, Weilu Gao, Minhan Lou, Katsumasa Yoshioka, Michael J. Manfra & Junichiro Kono 

Nature Photonics **12**, 324–329 (2018) | [Cite this article](#)

Cavity-mediated thermal control of metal-to-insulator transition in 1T-TaS₂

Giacomo Jarc, Shahla Yasmin Mathengattil, Angela Montanaro, Francesca Giusti, Enrico Maria Rigoni, Rudi Sergo, Francesca Fassioli, Stephan Winnerl, Simone Dal Zilio, Dragan Mihailovic, Peter Prelovšek, Martin Eckstein & Daniele Fausti 

Nature **622**, 487–492 (2023) | [Cite this article](#)

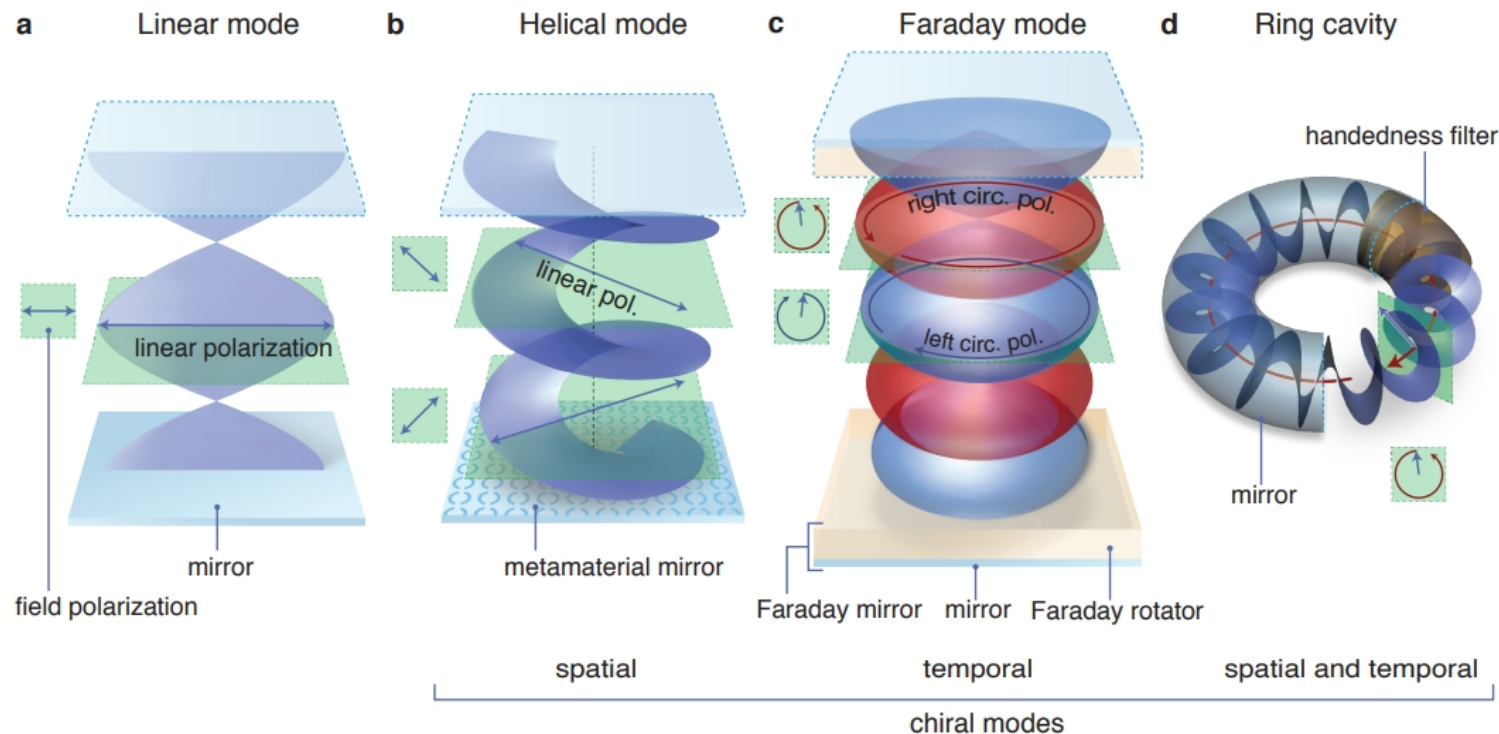
[Submitted on 31 Jul 2024 (v1), last revised 23 Apr 2025 (this version, v3)]

Cavity-Driven Attractive Interactions in Quantum Materials

F. Helmrich, H. S. Adlong, I. Khanonkin, M. Kroner, G. Scalari, J. Faist, A. Imamoglu, T. F. Nova

Design cavities with specific symmetries

2



SCIENCE ADVANCES | RESEARCH ARTICLE

PHYSICS

Chiral flat-band optical cavity with atomically thin mirrors

Daniel G. Suárez-Forero^{1†}, Ruihao Ni^{2†}, Supratik Sarkar^{1†}, Mahmoud Jalali Mehrabad^{1†}, Erik Mechtel¹, Valery Simonyan¹, Andrey Grankin¹, Kenji Watanabe³, Takashi Taniguchi⁴, Suji Park⁵, Houk Jang⁵, Mohammad Hafezi^{1*}, You Zhou^{2,6*}

nature communications



Article

<https://doi.org/10.1038/s41467-025-60335-x>

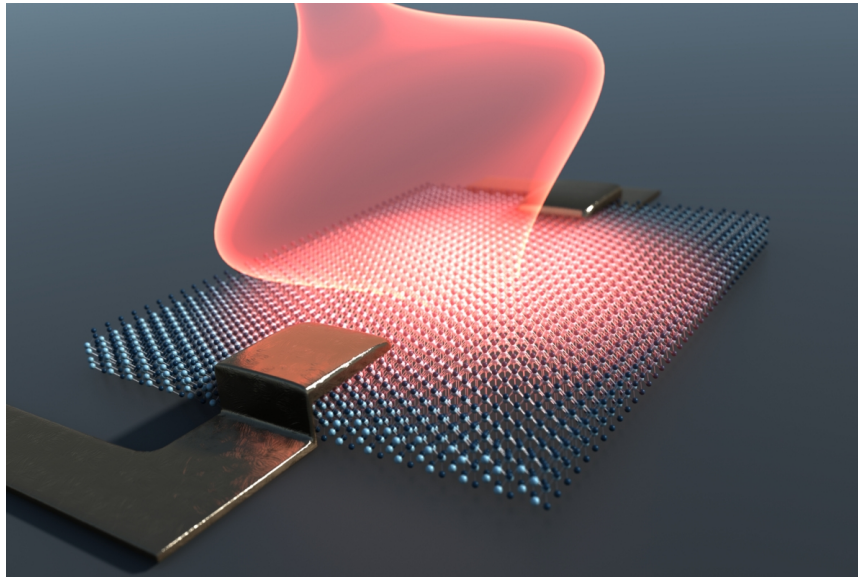
Terahertz chiral photonic-crystal cavities for Dirac gap engineering in graphene

Received: 28 October 2024

Accepted: 16 May 2025

Fuyang Tay^{1,2}, Stephen Sanders¹, Andrey Baydin^{1,3,4}, Zhigang Song⁵, Davis M. Welakuh⁶, Alessandro Alabastri^{1,3,4}, Vasil Rokaj^{7,8,9}, Ceren B. Dag^{7,8,10} & Junichiro Kono^{1,3,4,11,12}

Floquet engineering as a motivation



Main idea: produce a time dependent Hamiltonian (applying a laser or any time dependent variation of some parameter) so that

$$\hat{H}(t) = \hat{H}(t + T)$$

This introduces a non-trivial 'effective' behavior of the system

Note: the time dependent field is classical
(a bosonic degree of freedom in a coherent state)

Floquet theory

Equivalent to Bloch theorem for periodic time dependent systems

Bloch (space)

$$\hat{\mathcal{H}}(\mathbf{r} + \mathbf{R}) = \hat{\mathcal{H}}(\mathbf{r})$$

$$\Rightarrow \psi_{\mathbf{k}}(\mathbf{r}) = e^{i \mathbf{k} \cdot \mathbf{r}} \phi_{\mathbf{k}}(\mathbf{r})$$

quasi-momentum

Bloch function

$$\phi_{\mathbf{k}}(\mathbf{r} + \mathbf{R}) = \phi_{\mathbf{k}}(\mathbf{r})$$

$k \in \left[-\frac{\pi}{a}, \frac{\pi}{a}\right]$ (Brillouin zone)
or extended zone picture

Floquet (time)

$$\hat{\mathcal{H}}(t + T) = \hat{\mathcal{H}}(t)$$

$$\Rightarrow |\psi_{\alpha}(t)\rangle = e^{-i \varepsilon_{\alpha} t / \hbar} |\phi_{\alpha}(t)\rangle$$

quasi-energy

Floquet eigenvector

$$|\phi_{\alpha}(t + T)\rangle = |\phi_{\alpha}(t)\rangle$$

$\varepsilon_{\alpha} \in \left[-\frac{\hbar\Omega}{2}, \frac{\hbar\Omega}{2}\right]$ (Floquet BZ)
or extended zone picture

Shirley, Phys. Rev (1965); Sambe, PRA (1973).

Floquet in a nutshell: extended zone picture

$$\hat{H}(t) = \hat{H}(t + T)$$

Time periodic Hamiltonian

Floquet Theorem $\rightarrow$

$$|\psi_\alpha(t)\rangle = e^{-i\varepsilon_\alpha t} |u_\alpha(t)\rangle$$

ε_α : quasienergy

$|u_\alpha(t)\rangle = |u_\alpha(t + T)\rangle$: Floquet modes

$$(\hat{H}(t) - i\partial_t)|u_\alpha(t)\rangle = \varepsilon_\alpha |u_\alpha(t)\rangle$$

Mapping to a static **Floquet Hamiltonian**:

$$\hat{H}(t) = \sum_n e^{-in\Omega t} H_n$$

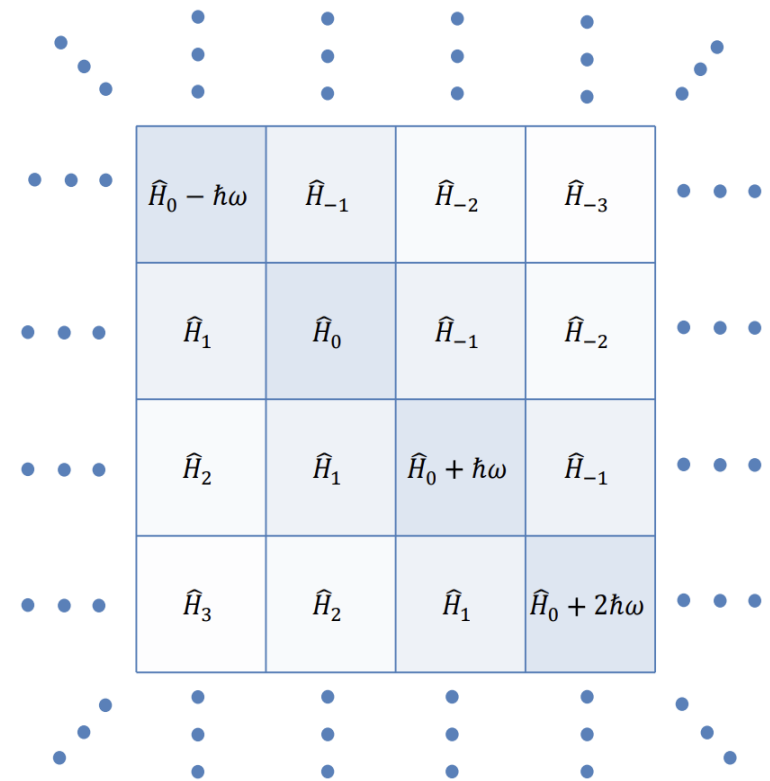
$$|u_\alpha(t)\rangle = \sum_n e^{-in\Omega t} |u_\alpha^{(n)}\rangle$$

$$|u_\alpha\rangle\rangle =$$

$$\begin{pmatrix} \vdots \\ |u_\alpha^1\rangle \\ |u_\alpha^0\rangle \\ |u_\alpha^{-1}\rangle \\ \vdots \end{pmatrix}$$

$$\hat{H}^F |u_\alpha\rangle\rangle = \varepsilon_\alpha |u_\alpha\rangle\rangle$$

$$\hat{H}^F =$$



J. H. Shirley, Phys. Rev. **138**, B979 (1965) $\omega \rightarrow \Omega$
 H. Sambe, PRA **7**, 2203 (1973)
 Eckardt & Anisimovas, New J. Phys. **17** 093039 (2015)

Effective Hamiltonian & micromotion

Floquet Theorem:

$$\exists \hat{R}(t), \hat{H}_{\text{eff}} = \hat{R}^\dagger(t) \hat{H}(t) \hat{R}(t) - i \hat{R}^\dagger(t) \partial_t \hat{R}(t)$$

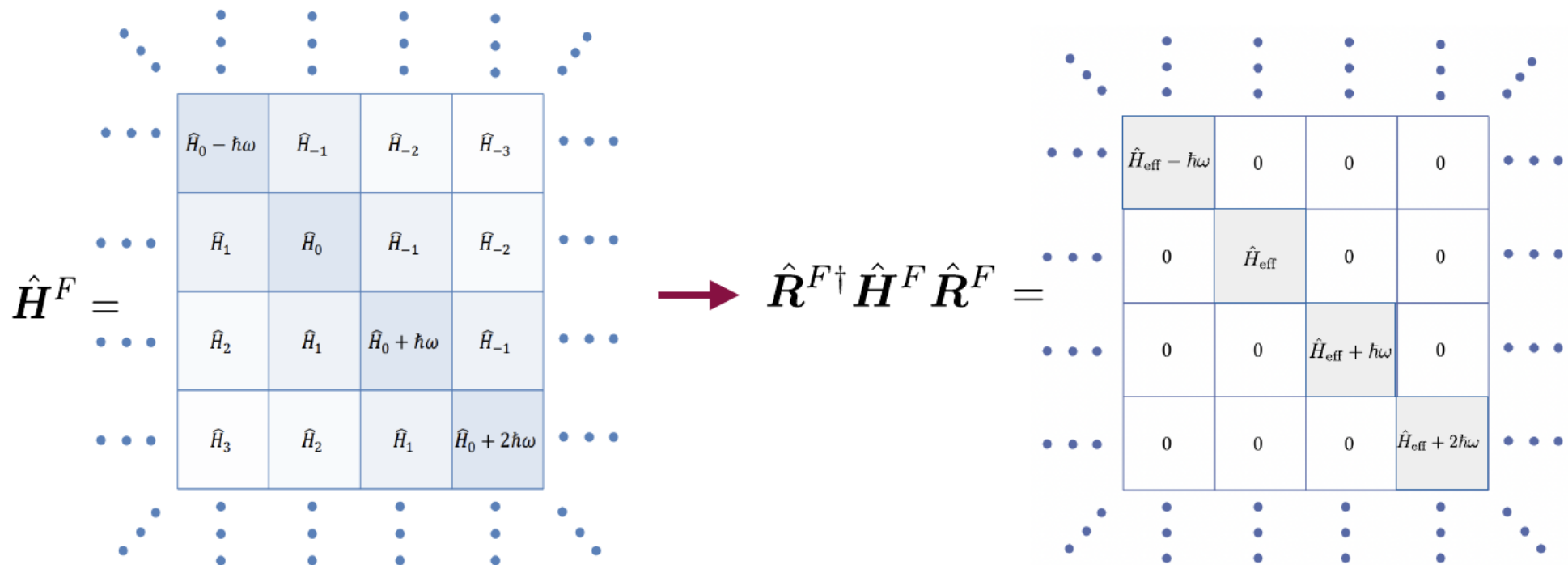
$$\hat{U}(t, t') = \hat{R}(t) e^{-i \hat{H}_{\text{eff}}(t-t')} \hat{R}^\dagger(t')$$

Goldman & Dalibard,
PRX **4**, 031027 (2014)

Eckardt & Anisimovas, New
J. Phys. **17** 093039 (2015)

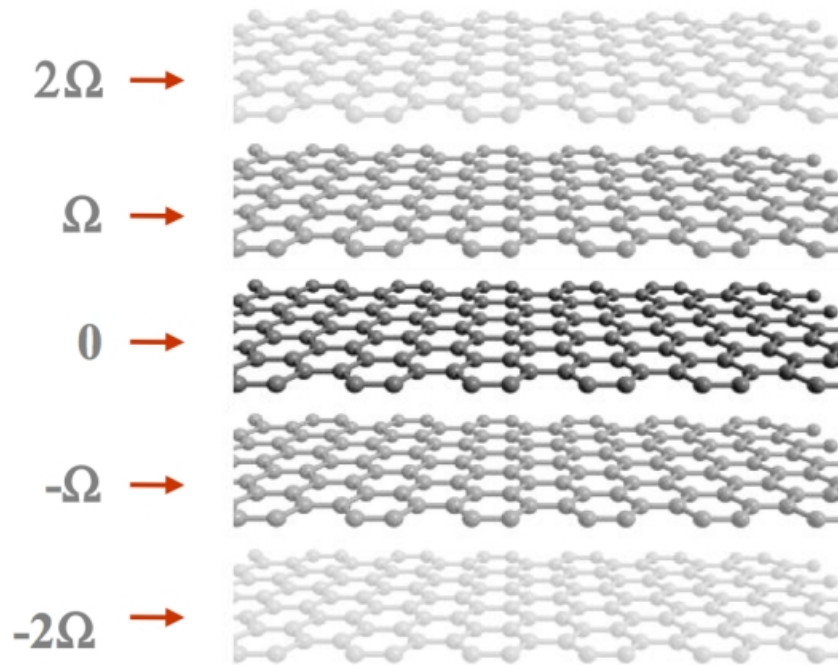
$\hat{H}_{\text{eff}}$: time-independent effective Hamiltonian

$\hat{R}(t)$: micro-motion operator, $\hat{R}(t) = \hat{R}(t + T)$

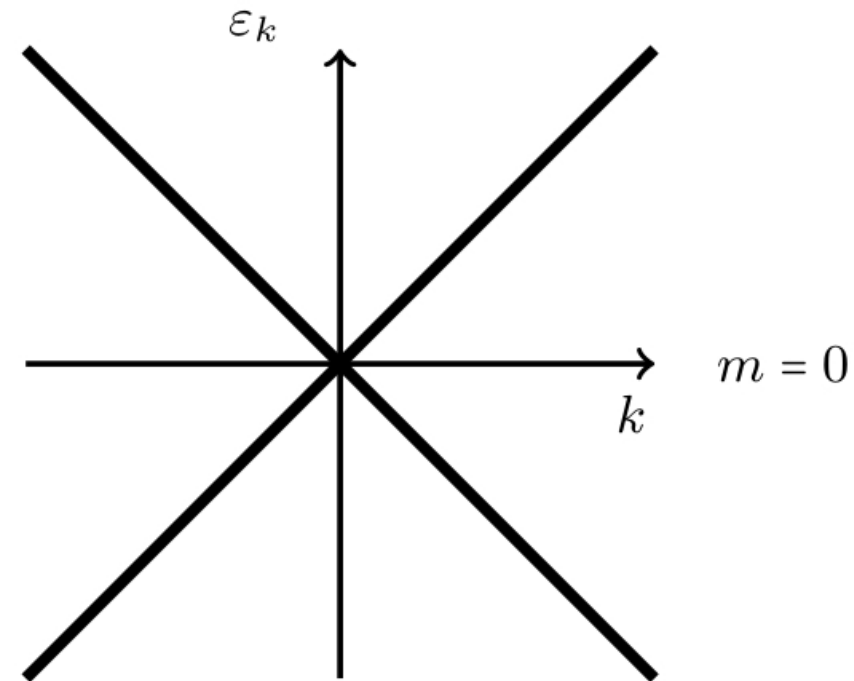


Floquet theory

Physical interpretation: 'electron $\otimes$ photons'

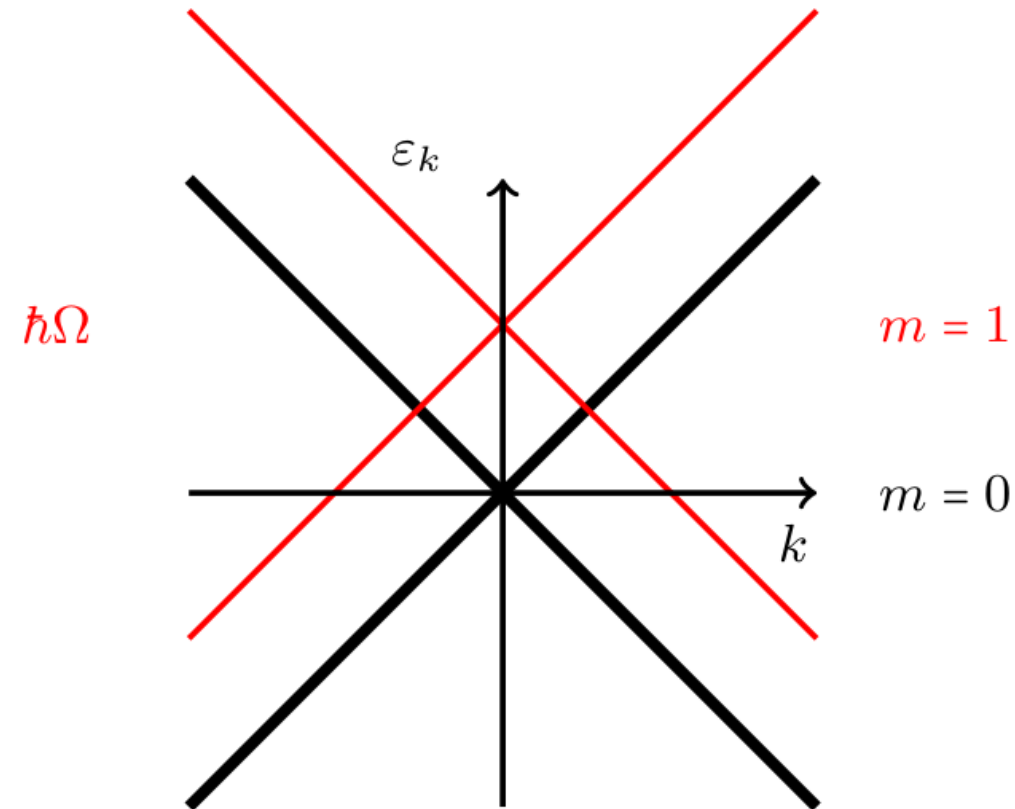
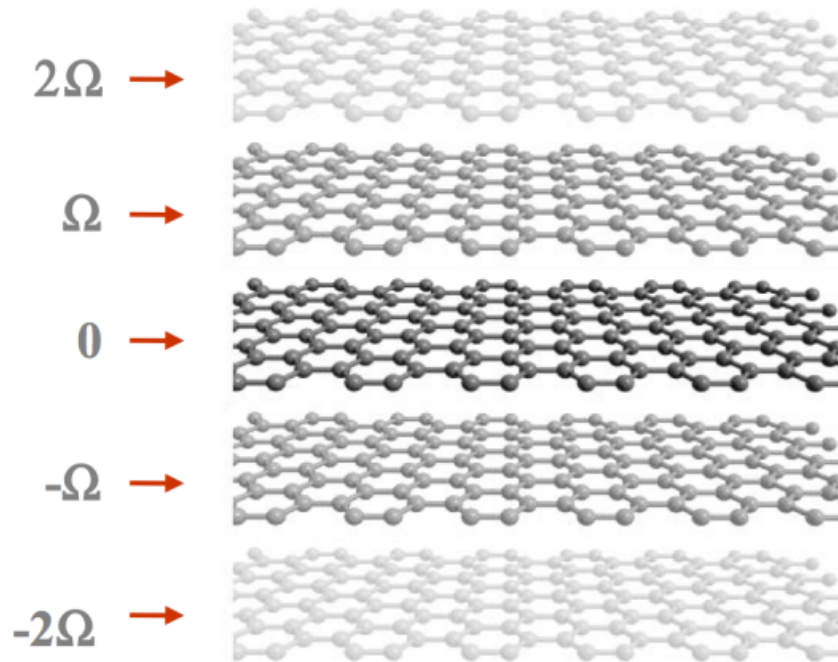


- Copies of the original system coupled by EM field
- Different number of photons



Floquet theory

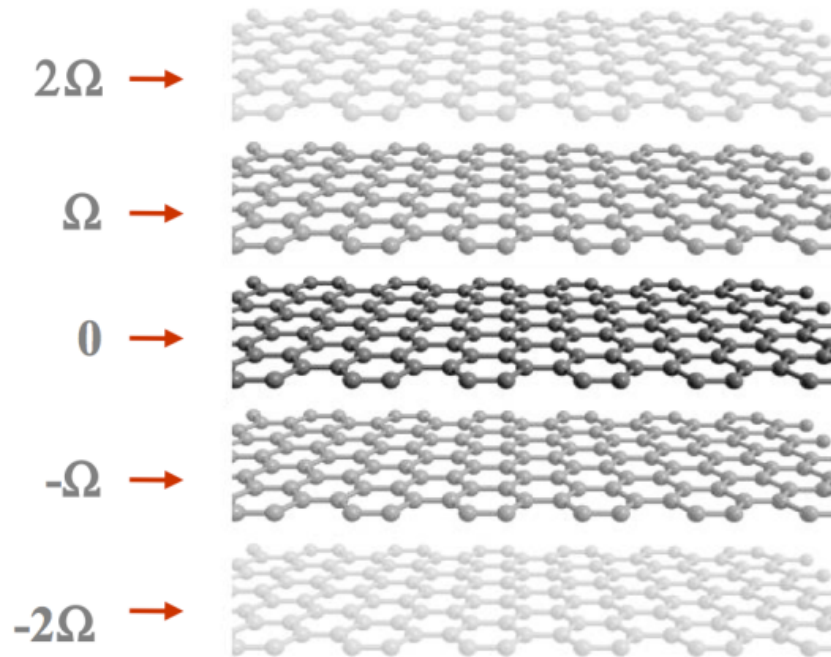
Physical interpretation: 'electron $\otimes$ photons'



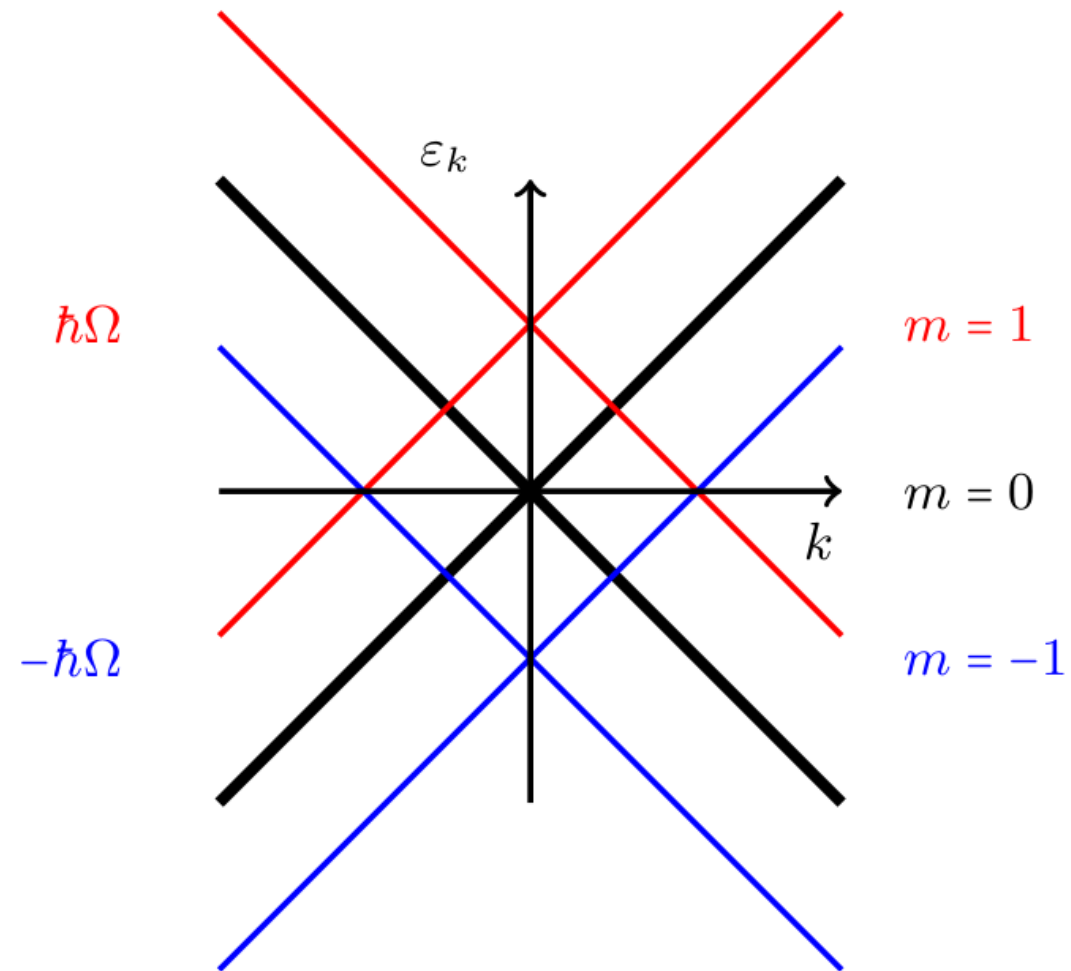
- Copies of the original system coupled by EM field
- Different number of photons

Floquet theory

Physical interpretation: 'electron $\otimes$ photons'

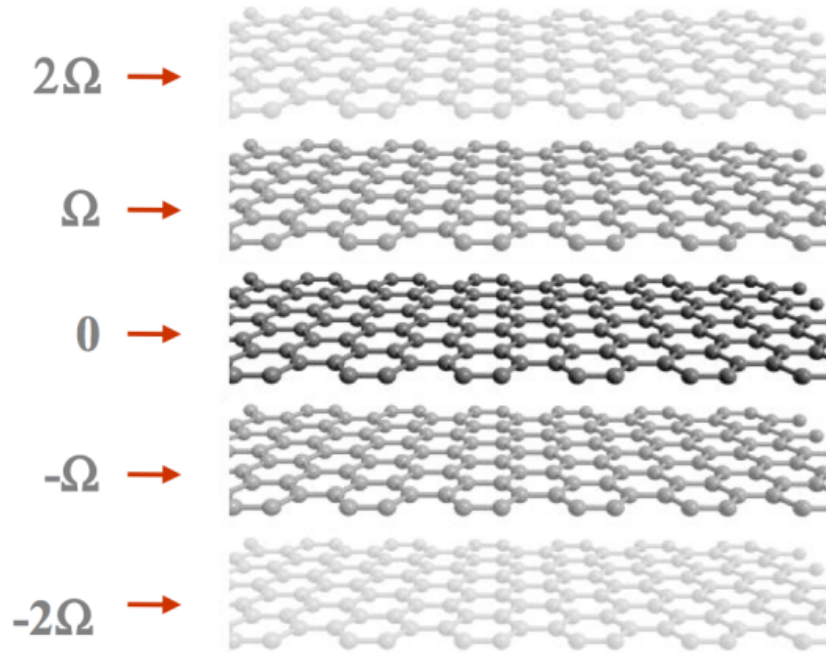


- Copies of the original system coupled by EM field
- Different number of photons

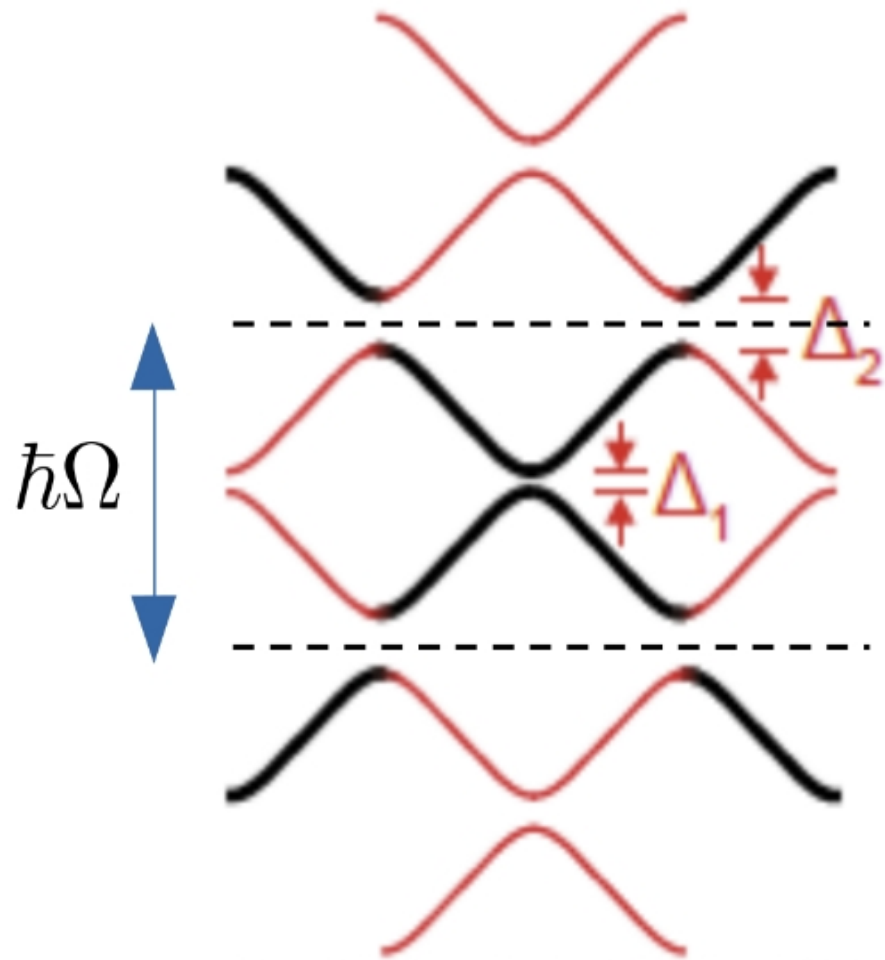


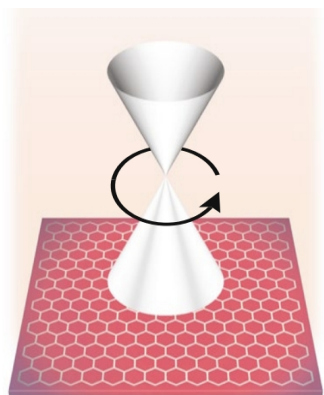
Floquet theory

Physical interpretation: 'electron $\otimes$ photons'

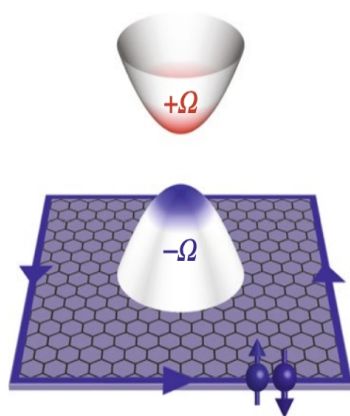


- Copies of the original system coupled by EM field
- Different number of photons



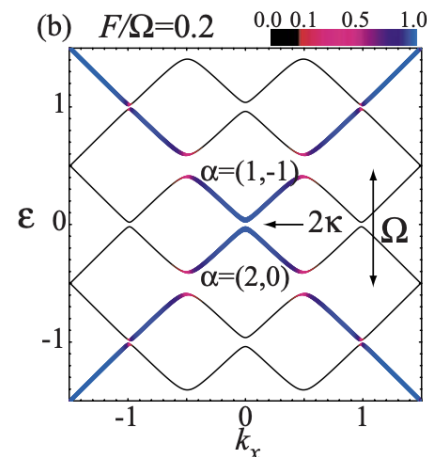


Time-dependent picture

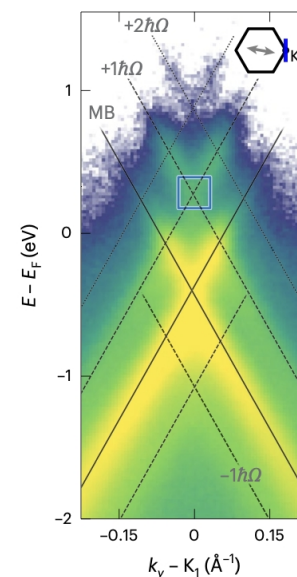
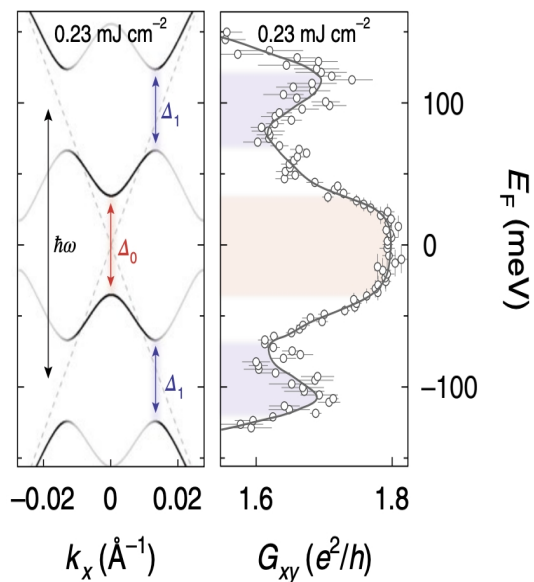
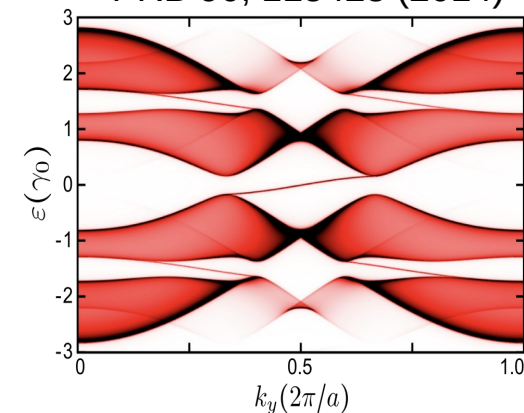


Effective Floquet picture

Oka & Aoki
PRB **79**, 081406R (2009)



Usaj, Perez-Piskunow,
Foa Torres & Balseiro,
PRB **90**, 115423 (2014)

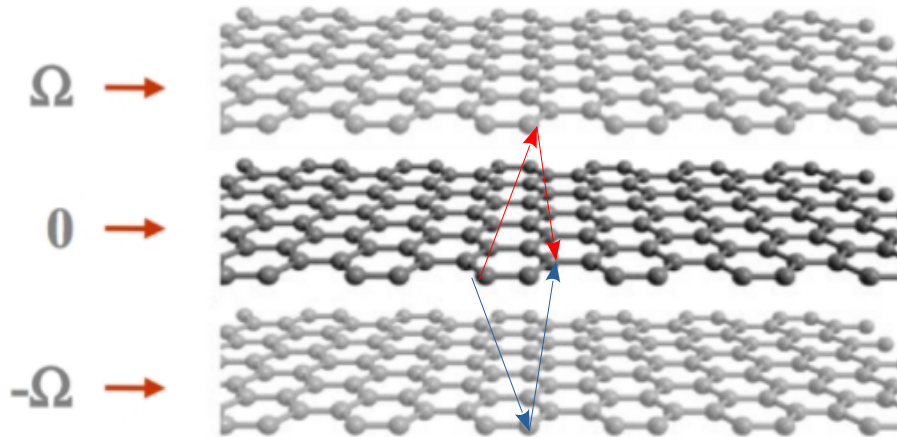
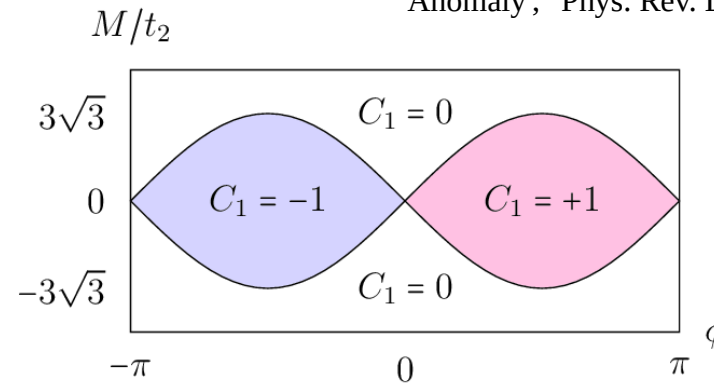
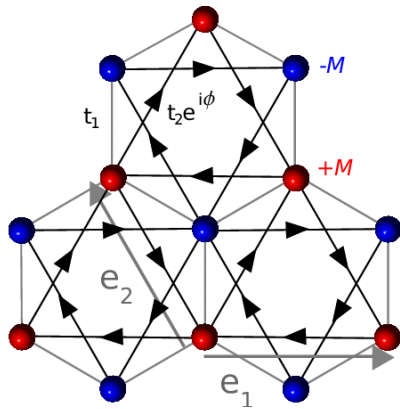


Observation of Floquet–Bloch states in monolayer
graphene, Choi et al, Nature Physics (2025)

Observation of Floquet states in graphene, Merboldt
et al, Nature Physics (2025)

Effective Haldane model

F. D. M. Haldane, "Model for a Quantum Hall Effect without Landau Levels: Condensed-Matter Realization of the 'Parity Anomaly'," Phys. Rev. Lett. 61, 2015 (1988)



$$H_{\text{eff}} = H_0 + \frac{1}{\Omega} \sum_{n \neq 0} \frac{[H_n, H_{-n}]}{n}$$

Circularly polarized laser

Can we make something similar using vacuum fluctuations?

Why? Heating, ground state physics, entanglement, correlations... for free...

Quantum limit of the Floquet systems

Box 1: Floquet and quantum light-matter dressing at a glance

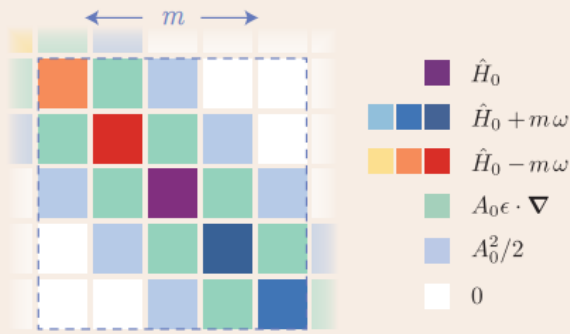
Floquet materials engineering. The Floquet formalism decomposes the dynamics of a periodically driven system into static states that are harmonics of the driving frequency ω and constant phases E_α with the ansatz

$$|\Psi_\alpha(t)\rangle = e^{-iE_\alpha t} \sum_{m=-\infty}^{\infty} e^{im\omega t} |\psi_m^\alpha\rangle$$

The static states ψ_m^α are the components of the dressed state for the harmonic m . In the basis of harmonics the time-dependent Schrödinger equation reads

$$\begin{aligned} \{E_\alpha - m\omega - \hat{H}_0\} \psi_m^\alpha(\mathbf{r}) \\ = A_0 \epsilon \cdot \nabla (\psi_{m-1}^\alpha(\mathbf{r}) - \psi_{m+1}^\alpha(\mathbf{r})) \\ - \frac{1}{2} A_0^2 (\psi_{m-2}^\alpha(\mathbf{r}) + \psi_{m+2}^\alpha(\mathbf{r})) \end{aligned}$$

which can be visualized as the matrix structure:



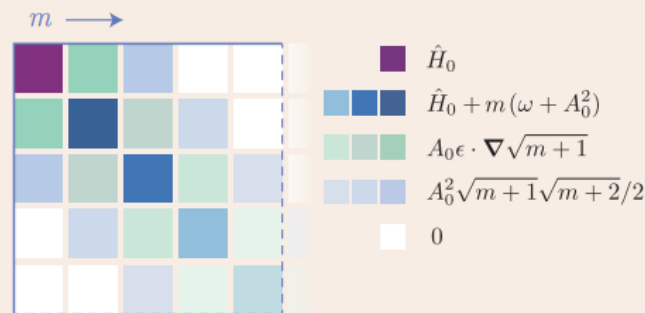
QED materials engineering. The fundamental theory for the description of light-matter coupling is quantum electrodynamics (QED), where both light and matter are described in terms of quantized particles. The coupled electron-photon state can be written as a dressed state

$$|\Psi_\alpha\rangle = \sum_{m=0}^{\infty} |m\rangle |\psi_m^\alpha\rangle$$

where $|m\rangle$ are coherently shifted Fock number states [22], ψ_m^α the electronic components. In the basis of these number states, the QED Hamiltonian reads

$$\begin{aligned} \{E_\alpha - (\omega + A_0^2) \left(m + \frac{1}{2}\right) - \hat{H}_0\} \psi_m^\alpha(\mathbf{r}) \\ = A_0 \epsilon \cdot \nabla (\sqrt{m} \psi_{m-1}^\alpha(\mathbf{r}) - \sqrt{m+1} \psi_{m+1}^\alpha(\mathbf{r})) \\ - \frac{1}{2} A_0^2 (\sqrt{m} \sqrt{m-1} \psi_{m-2}^\alpha(\mathbf{r}) + \sqrt{m+1} \sqrt{m+2} \psi_{m+2}^\alpha(\mathbf{r})) \end{aligned}$$

which can be visualized as the matrix structure:



- Hilbert space is bounded
- Vacuum fluctuations
- Correlations
- Access to the photon sector ('Floquet replicas')

Engineering quantum materials with chiral optical cavities

Hannes Hübener [✉](#), Umberto De Giovannini [✉](#), Christian Schäfer, Johan Andberger, Michael Ruggenthaler, Jerome Faist & Angel Rubio [✉](#)

Nature Materials **20**, 438–442 (2021) | [Cite this article](#)

The quantum Rabi model

$$H_a = \frac{\hbar\omega_a}{2}\sigma_z$$

$$H_c = \hbar\omega_c a^\dagger a$$

$$\left(\begin{array}{c} \text{---} |e\rangle \\ \text{---} |g\rangle \end{array} \right)$$

$$\sigma_z = |e\rangle\langle e| - |g\rangle\langle g|$$

$$H_{\text{int}} = -\mathbf{d} \cdot \mathbf{E} = -\hbar g(\sigma_+ + \sigma_-)(a^\dagger + a)$$

$$\sigma_+ = |e\rangle\langle g|$$

$$\sigma_- = |g\rangle\langle e|$$

Dipolar approximation

$$\lambda \gg a_0$$

$$\hbar g = \sqrt{\frac{2\pi\hbar\omega_c}{V}} \langle e | \mathbf{d} \cdot \boldsymbol{\varepsilon} | g \rangle$$

$$\mathbf{E} = \sqrt{\frac{2\pi\hbar\omega_c}{V}} (a^\dagger + a) \boldsymbol{\varepsilon}$$

Rabi coupling

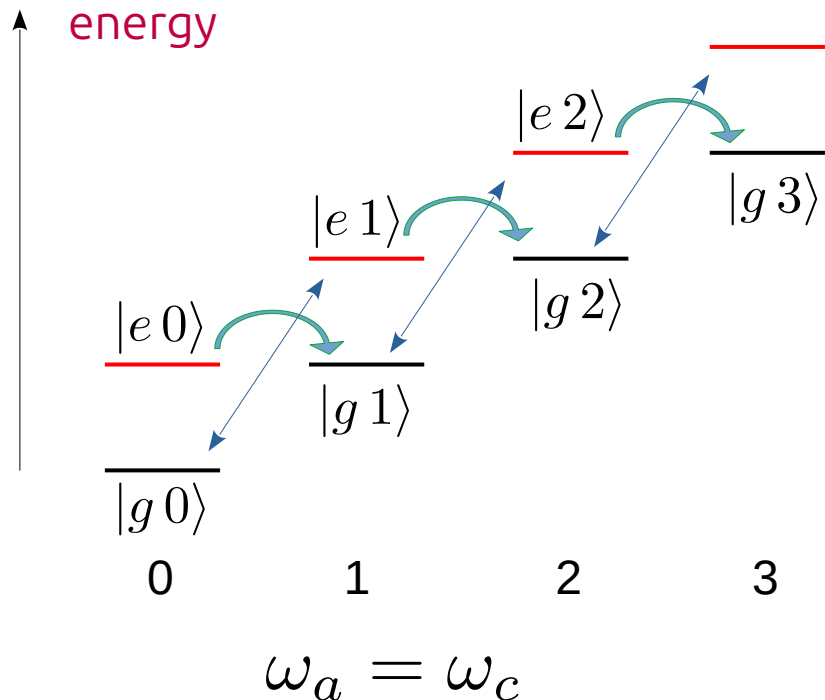
Electric field operator

The quantum Rabi model

$$H = H_0 + H_{\text{int}}$$

$$H_0 = \hbar\omega_c a^\dagger a + \frac{\hbar\omega_a}{2} \sigma_z$$

$$H_{\text{int}} = -\hbar g(\sigma_+ + \sigma_-)(a^\dagger + a) = \underbrace{-\hbar g(\sigma_+ a + \sigma_- a^\dagger)}_{\text{Resonant emission/absorption}} - \underbrace{\hbar g(\sigma_- a + \sigma_+ a^\dagger)}_{\text{Non-resonant term (counter-rotating)}}$$



$$H_{\text{JC}} = H_0 - \hbar g(\sigma_+ a + \sigma_- a^\dagger)$$

Jaynes-Cummings model

The JC model as a RWA

$$H_{\text{int}} = -\hbar g(\sigma_+ a + \sigma_- a^\dagger) - \hbar g(\sigma_- a + \sigma_+ a^\dagger)$$

Interaction picture $\rightarrow$ $a(t) = a e^{-i\omega_c t}$ $\sigma_-(t) = \sigma_- e^{-i\omega_a t}$

$$\sigma_+ a \quad e^{+i(\omega_a - \omega_c)t} \quad \text{Slow}$$

$$\sigma_- a^\dagger \quad e^{-i(\omega_a - \omega_c)t} \quad \text{Slow}$$

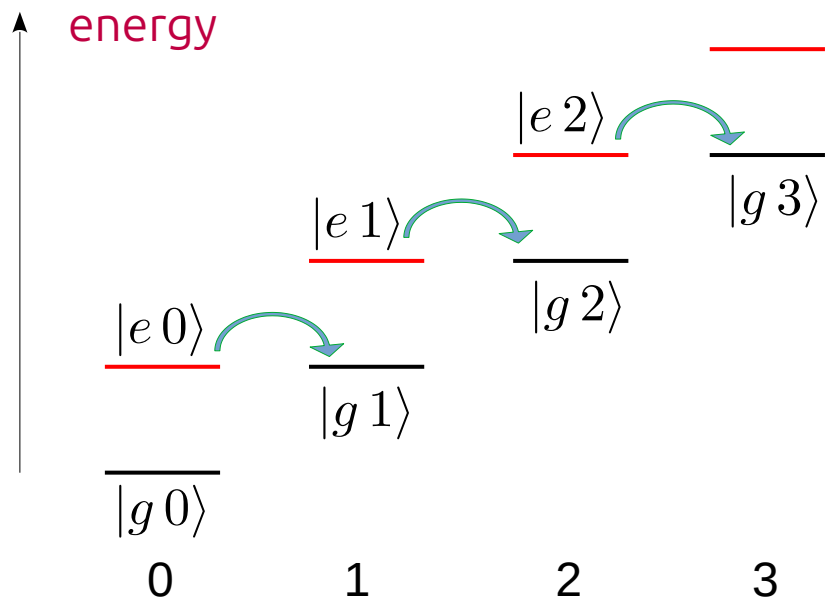
$$\sigma_+ a^\dagger \quad e^{+i(\omega_a + \omega_c)t} \quad \text{Fast}$$

$$\sigma_- a \quad e^{-i(\omega_a + \omega_c)t} \quad \text{Fast}$$

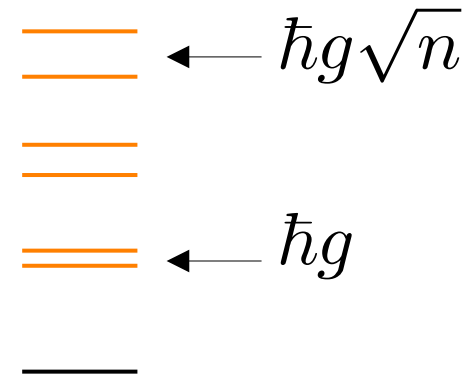
Counter-rotating
 $\rightarrow$ Average to zero $\rightarrow$ drop

Spectrum JC model

$$H_{\text{JC}} = H_0 - \hbar g (\sigma_+ a + \sigma_- a^\dagger)$$



$$g \ll \omega_a$$



$$|n\pm\rangle = \frac{1}{\sqrt{2}} (|e, (n-1)\rangle \pm |g, n\rangle)$$

For $n=1$

$$|\pm\rangle = \frac{1}{\sqrt{2}} (|e, 0\rangle \pm |g, 1\rangle)$$

$$E_{\pm} = \frac{\hbar\omega_a}{2} \pm \hbar g$$

Polariton (light and matter are entangled)
 → one single entity

Vacuum Rabi oscillations

$$|\psi(0)\rangle = |e\ 0\rangle$$

Initial state

$\Rightarrow$

$$|\psi(t)\rangle = (|+\rangle e^{-iE_+t/\hbar} + |-\rangle e^{-iE_-t/\hbar})/\sqrt{2}$$

Time evolution

$$P(t) = \sum_n |\langle g\ n|\psi(t)\rangle|^2 = |\langle g\ 1|\psi(t)\rangle|^2 = \sin^2(gt)$$

Probability to find the atom in the ground state oscillates
even if initially the cavity was empty

A cavity material example: Landau polaritons

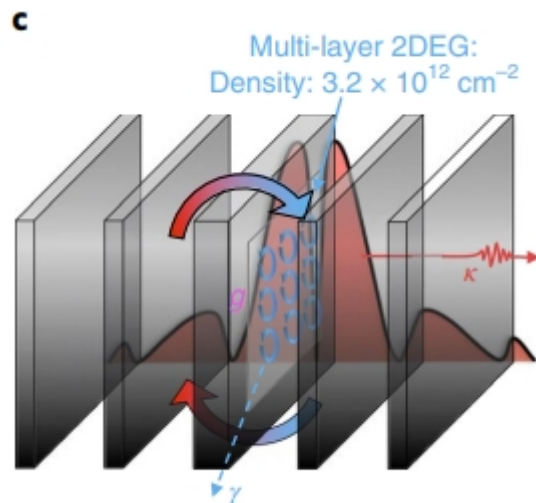
ARTICLES

<https://doi.org/10.1038/s41566-018-0153-0>

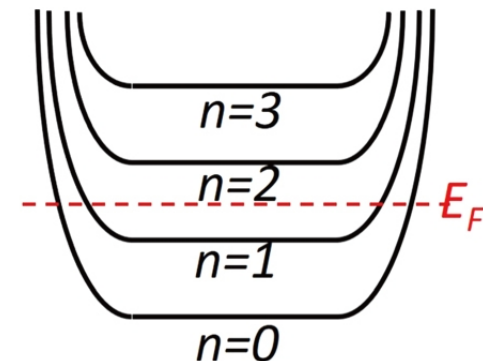
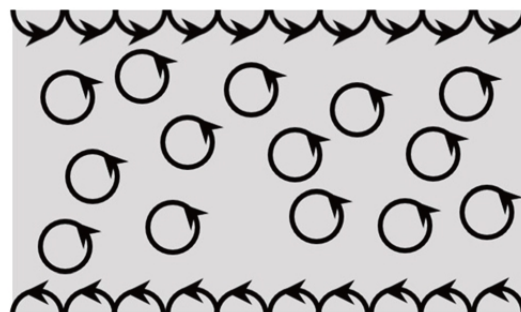
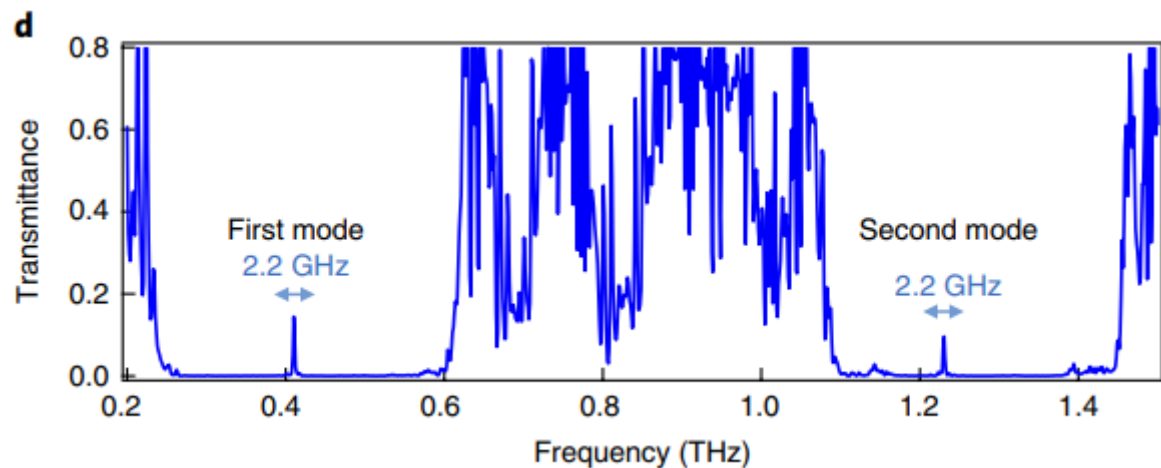
nature
photonics

Vacuum Bloch–Siegert shift in Landau polaritons with ultra-high cooperativity

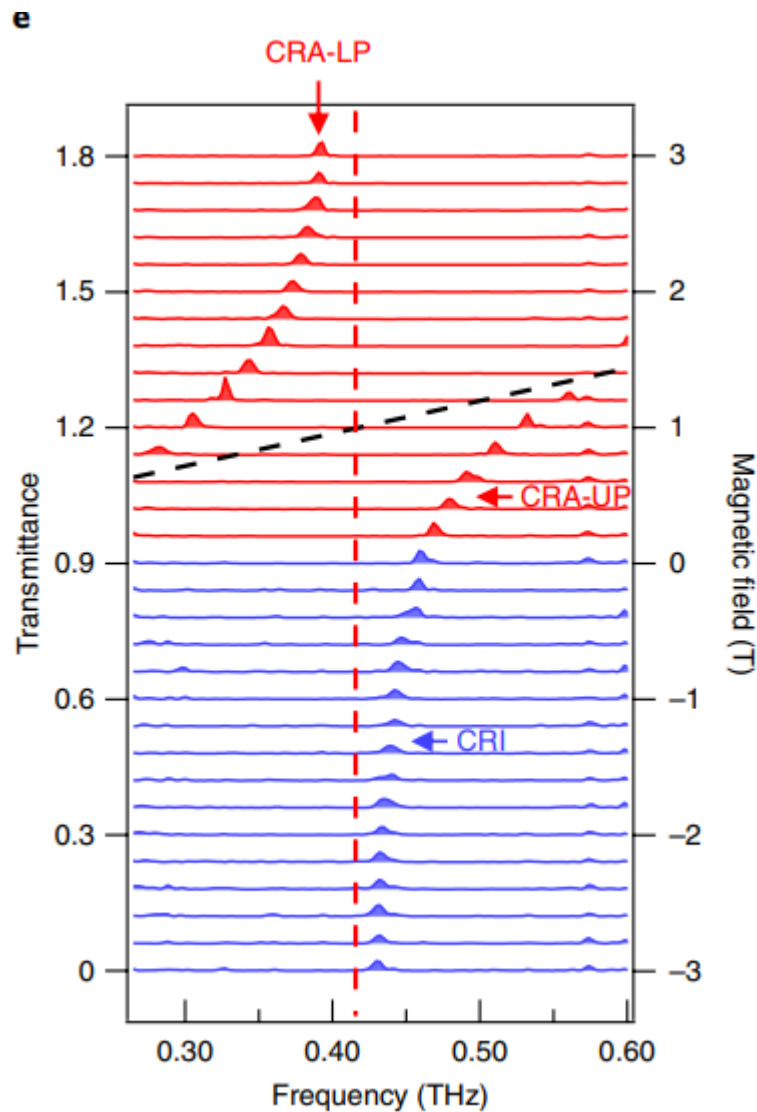
Xinwei Li¹, Motoaki Bamba^{2,3}, Qi Zhang⁴, Saeed Fallahi⁵, Geoff C. Gardner⁵, Weilu Gao¹, Minhan Lou¹, Katsumasa Yoshioka⁶, Michael J. Manfra^{5,7} and Junichiro Kono^{1,8,9*}



2DEG in
quantum Hall
regime

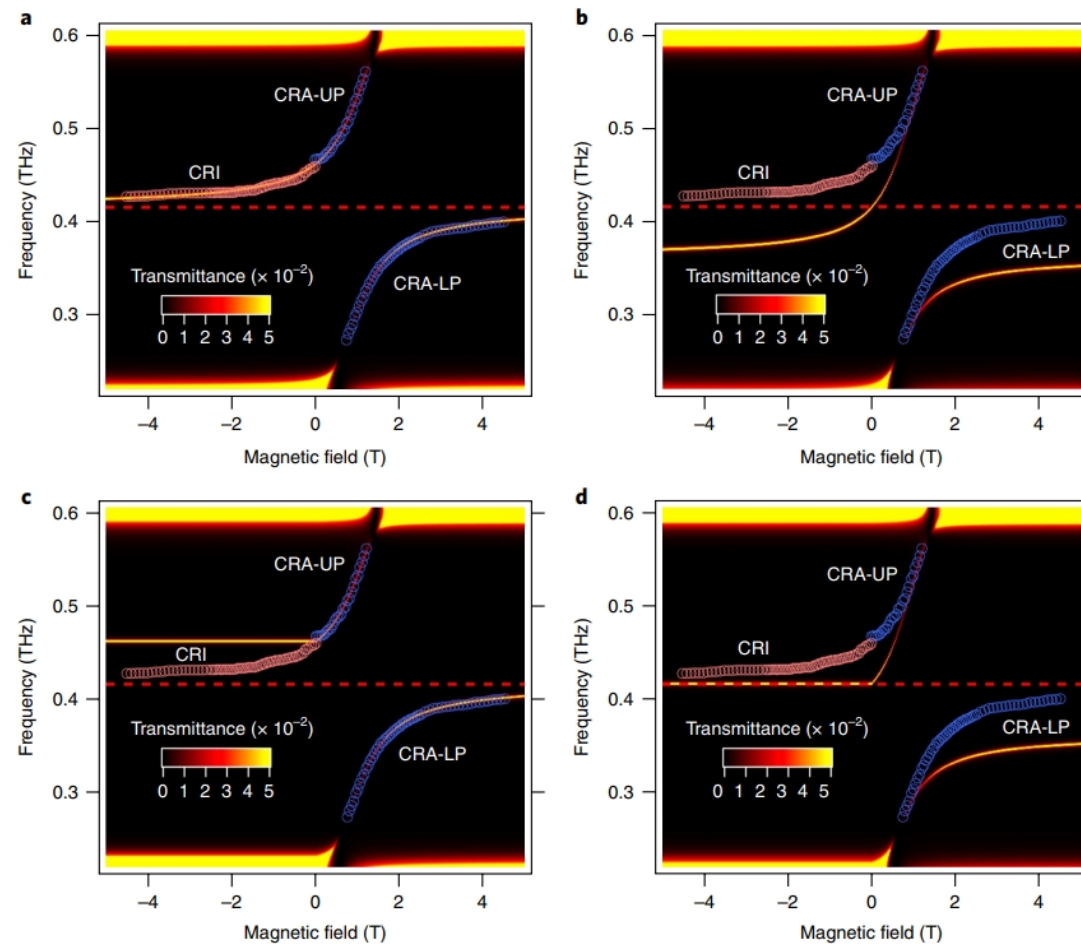


A cavity material example: Landau polaritons



$$g/\omega_c \sim 0.3$$

Ultra strong coupling



A couple of very simple examples

Quantum Floquet engineering with an exactly solvable tight-binding chain in a cavity

Christian J. Eckhardt ^{1,2,6}, Giacomo Passetti^{1,6}, Moustafa Othman³, Christoph Karrasch³, Fabio Cavaliere^{4,5}, Michael A. Sentef ² & Dante M. Kennes ^{1,2} 

COMMUNICATIONS PHYSICS | (2022)5:122

Controlling topological phases of matter with quantum light

Olesia Dmytruk¹  & Marco Schirò¹ 

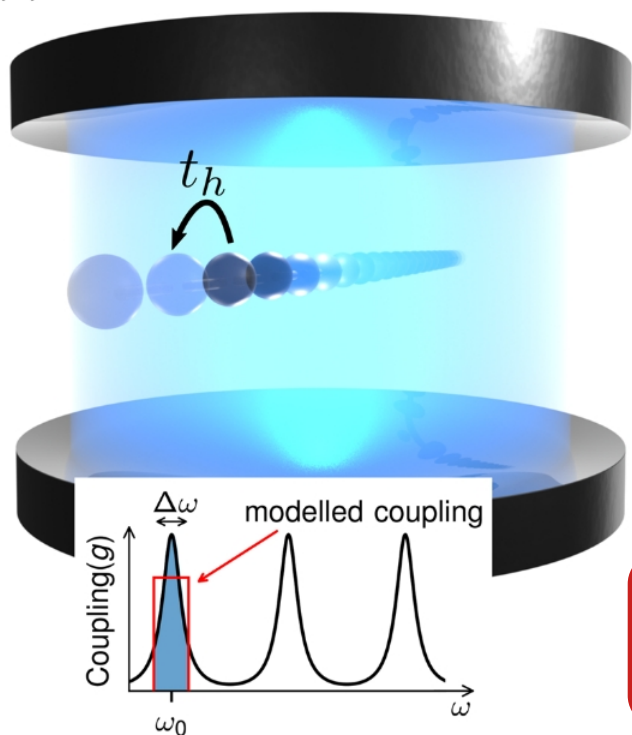
COMMUNICATIONS PHYSICS | (2022)5:271

Quantum Floquet engineering with an exactly solvable tight-binding chain in a cavity

COMMUNICATIONS PHYSICS | (2022)5:122

Christian J. Eckhardt^{1,2,6}, Giacomo Passetti^{1,6}, Moustafa Othman³, Christoph Karrasch³, Fabio Cavaliere^{4,5}, Michael A. Sentef² & Dante M. Kennes^{1,2}

(a)



$$H = \omega_0 \left(a^\dagger a + \frac{1}{2} \right) - \sum_{j=1}^L \left[t_h e^{-i \frac{g}{\sqrt{L}} (a^\dagger + a)} c_{j+1}^\dagger c_j + \text{h.c.} \right]$$

$$A = \frac{g}{\sqrt{L}} (a + a^\dagger)$$

Dipolar approximation

Peierls substitution
(delicate point)

$$H = \cos \left(\frac{g}{\sqrt{L}} (a + a^\dagger) \right) \mathcal{T} + \sin \left(\frac{g}{\sqrt{L}} (a + a^\dagger) \right) \mathcal{J} + \omega_0 \left(a^\dagger a + \frac{1}{2} \right)$$

Decoupled H operator

Current operator

$$\mathcal{T} := \sum_k -2t_h \cos(k) c_k^\dagger c_k =: \sum_k \varepsilon_k c_k^\dagger c_k$$

$$\mathcal{J} := \sum_k 2t_h \sin(k) c_k^\dagger c_k =: \sum_k v_k c_k^\dagger c_k$$

Note that

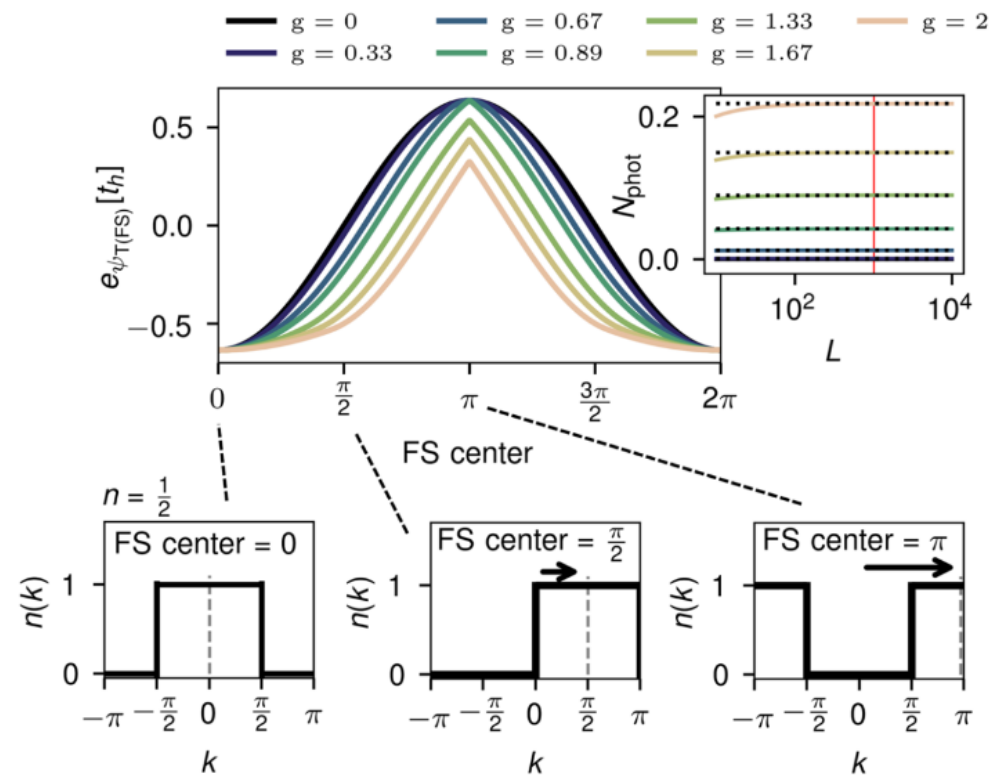
$$[\rho_k, H] = 0 \quad \text{for all } k \in \text{BZ}$$

$$\rho_k = c_k^\dagger c_k$$

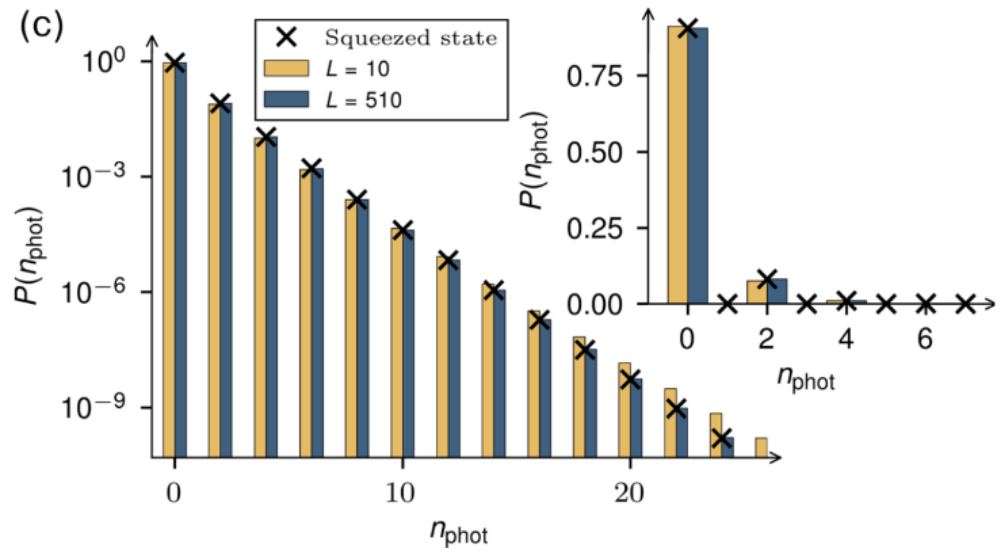
$$H|\Psi\rangle = E_\Psi|\Psi\rangle$$

$$|\Psi\rangle = |\phi\rangle_b \otimes |\psi\rangle_f$$

$$\rho = \frac{1}{L} \sum_k c_k^\dagger c_k$$



$$e_{\psi_{T(\text{FS})}} = \frac{E_{\psi_{T(\text{FS})}}}{L}$$



Photonic squeezed state

Why a squeezed state?

$$H = \cos\left(\frac{g}{\sqrt{L}}(a + a^\dagger)\right)\mathcal{T} + \sin\left(\frac{g}{\sqrt{L}}(a + a^\dagger)\right)\mathcal{J} + \omega_0\left(a^\dagger a + \frac{1}{2}\right)$$

$$\begin{aligned}\langle\Psi_{\text{GS}}|H|\Psi_{\text{GS}}\rangle &= \underbrace{\langle\Psi_{\text{GS}}|\omega_0\left(a^\dagger a + \frac{1}{2}\right)|\Psi_{\text{GS}}\rangle}_{\sim 1} + \underbrace{\langle\Psi_{\text{GS}}|\mathcal{T}|\Psi_{\text{GS}}\rangle}_{\sim L} \\ &\quad + \underbrace{\langle\Psi_{\text{GS}}|\frac{g}{\sqrt{L}}(a^\dagger + a)\mathcal{J}|\Psi_{\text{GS}}\rangle}_{\sim\sqrt{L}} \\ &\quad - \underbrace{\langle\Psi_{\text{GS}}|\frac{1}{2}\frac{g^2}{L}(a^\dagger + a)^2\mathcal{T}|\Psi_{\text{GS}}\rangle}_{\sim 1} + \mathcal{O}\left(\frac{1}{\sqrt{L}}\right).\end{aligned}$$

$$\tilde{\omega} = \omega_0 \sqrt{1 + 2\frac{g^2}{\omega_0} |t_{\text{GS}}|}$$

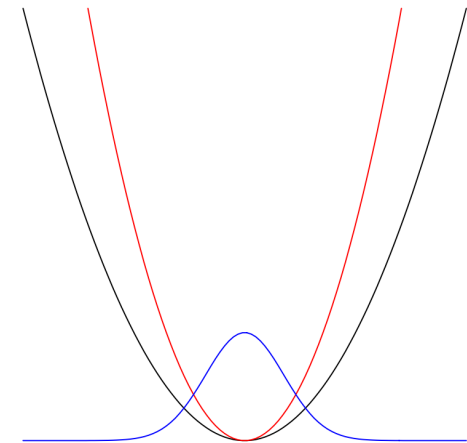
Renormalized frequency

$$|\phi_{\text{GS}}\rangle_b = e^{\frac{1}{2}(\zeta^* a^2 - \zeta (a^\dagger)^2)} |0\rangle$$

$$\zeta = \frac{1}{2} \ln\left(\frac{\tilde{\omega}}{\omega_0}\right)$$

$$t_{\text{GS}} = \frac{f\langle\Psi_{\text{GS}}|\mathcal{T}|\Psi_{\text{GS}}\rangle_f}{L} < 0,$$

$$j_{\text{GS}} = \frac{f\langle\Psi_{\text{GS}}|\mathcal{J}|\Psi_{\text{GS}}\rangle_f}{L} = 0,$$

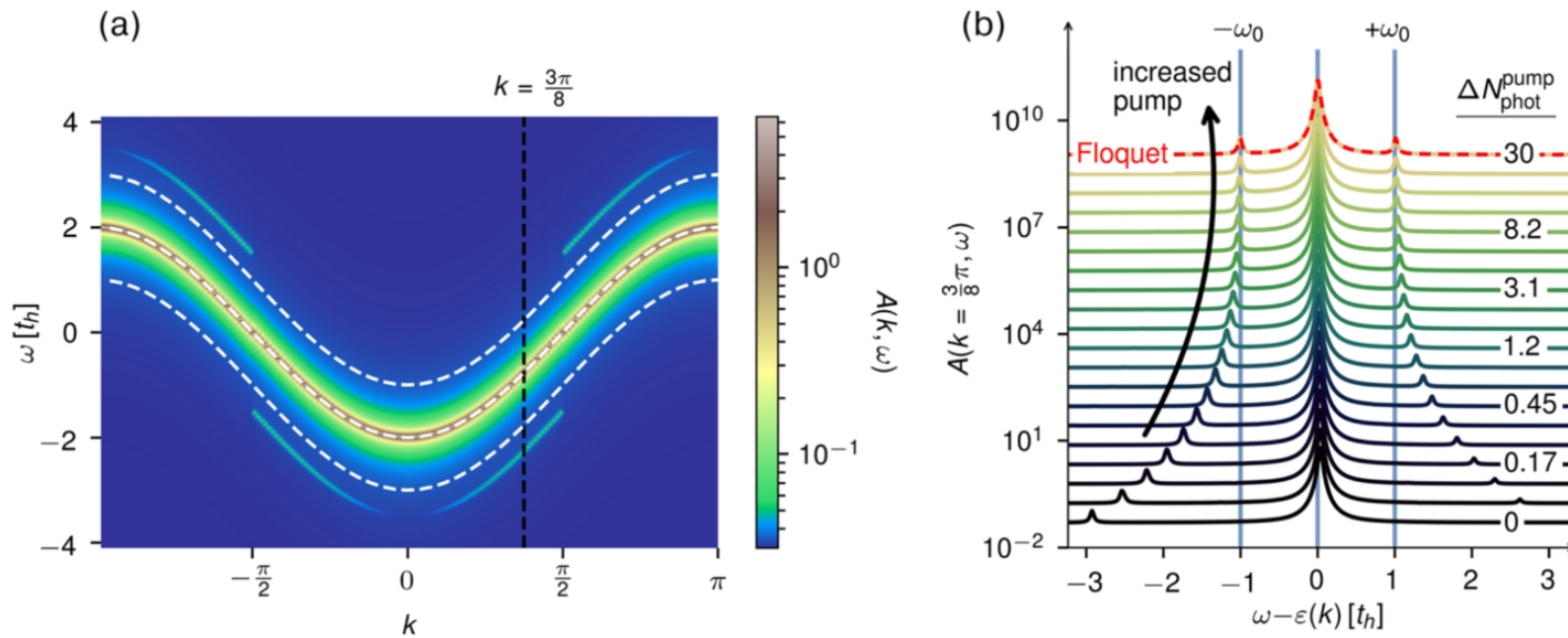


Recovering the Floquet limit?

$$|\alpha\rangle = e^{\alpha(a^\dagger - a)} |\phi_{\text{GS}}\rangle_b$$

$$|\alpha|^2 = \Delta N_{\text{phot}}^{\text{pump}}$$

Displaced GS



False superradiance transition

$$H^{1^{\text{st}}} = \omega_0 \left(a^\dagger a + \frac{1}{2} \right) + \mathcal{T} + \frac{g}{\sqrt{L}} (a^\dagger + a) \mathcal{J}$$

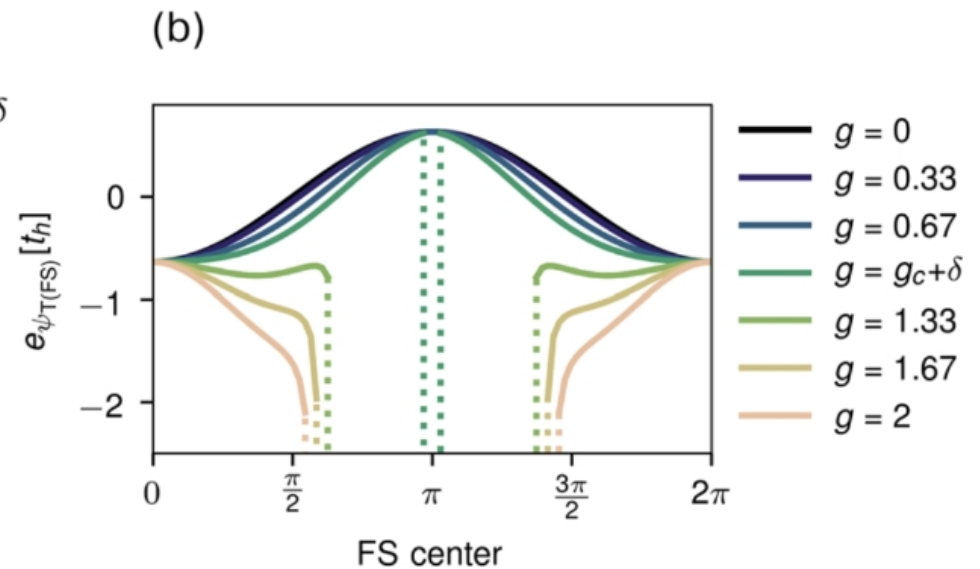
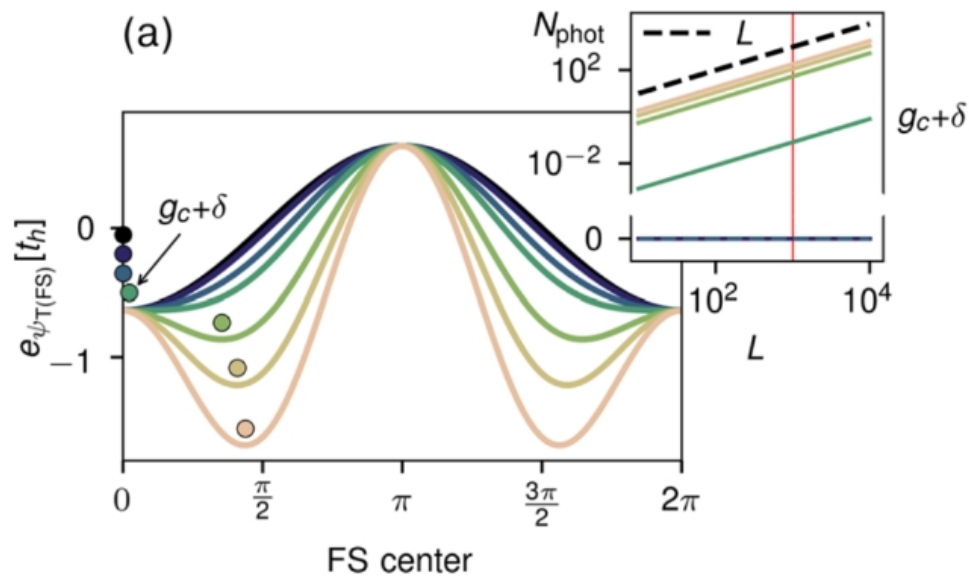
$$H^{2^{\text{nd}}} = \omega_0 \left(a^\dagger a + \frac{1}{2} \right) + \mathcal{T} + \frac{g}{\sqrt{L}} (a^\dagger + a) \mathcal{J} - \frac{1}{2} \frac{g^2}{L} (a^\dagger + a)^2 \mathcal{T}$$

$$g_c = \sqrt{\frac{\pi \omega_0}{4 t_h}}$$

$$\langle A \rangle = \frac{g \sqrt{L}}{\omega_0} j_{\text{GS}}$$

$$\langle A \rangle \sim \sqrt{N_{\text{emitter}}}$$

Wrong

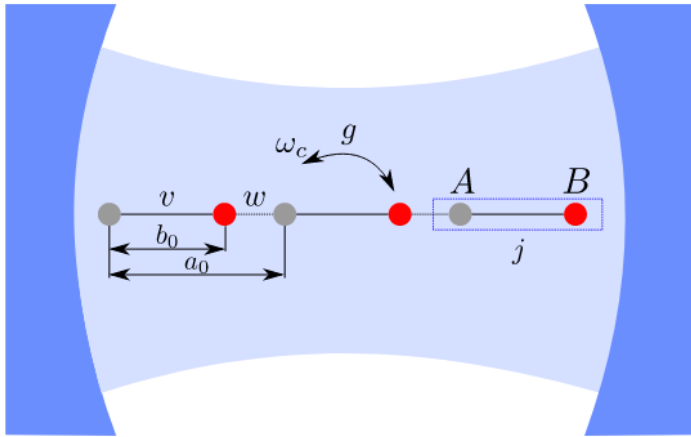


Keeping all orders of the Peierls phase (gauge invariance) is crucial

Controlling topological phases of matter with quantum light

COMMUNICATIONS PHYSICS | (2022)5:271

Olesia Dmytruk¹ & Marco Schirò¹



$$H = v \sum_{j=1}^L \left(e^{i \frac{g}{\sqrt{L}} b_0 (a + a^\dagger)} c_{jA}^\dagger c_{jB} + \text{h.c.} \right) - w \sum_{j=1}^{L-1} \left(e^{-i \frac{g}{\sqrt{L}} (1-b_0) (a + a^\dagger)} c_{j+1,A}^\dagger c_{jB} + \text{h.c.} \right) + \omega_c \left(a^\dagger a + \frac{1}{2} \right).$$

$$|\Psi\rangle = |\psi\rangle |\phi\rangle$$

Mean field approximation

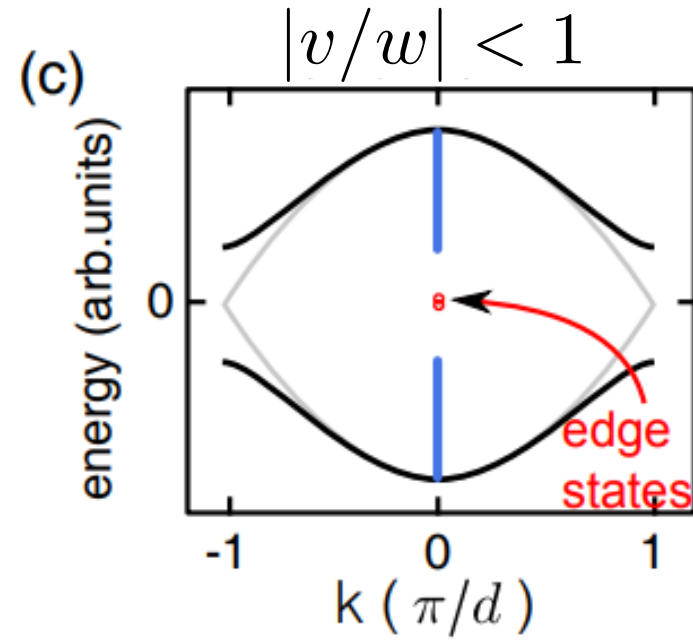
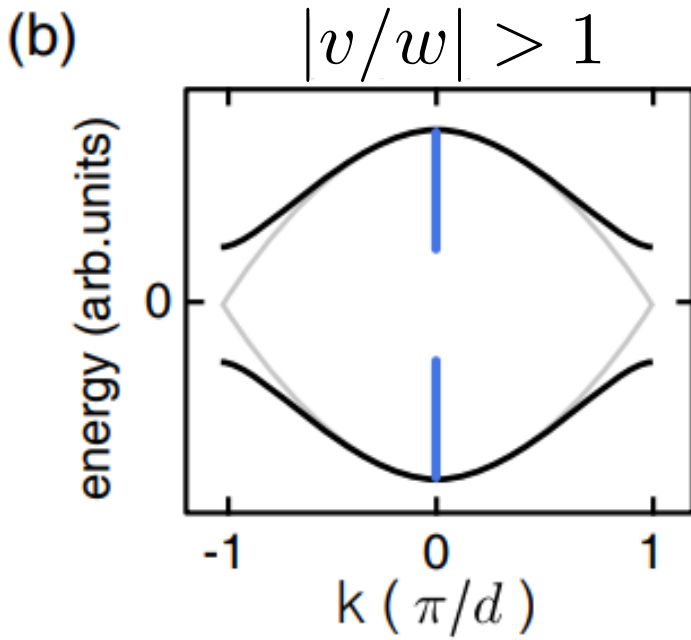
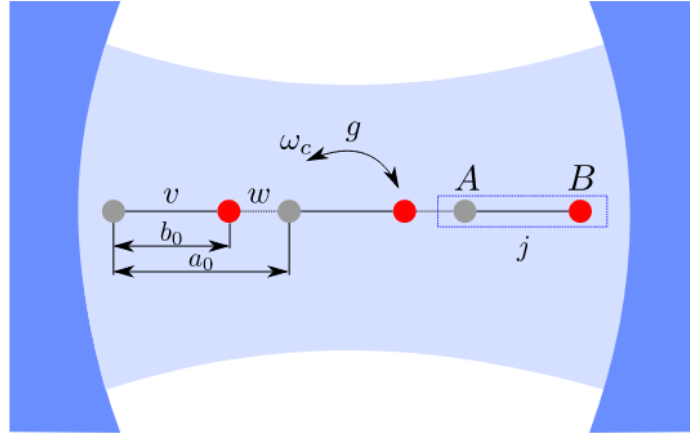
$$H_{\text{el}}^{\text{mf}} = \langle \phi | H | \phi \rangle = \sum_{j=1}^L \left(\tilde{v} c_{jA}^\dagger c_{jB} + \text{h.c.} \right) - \sum_{j=1}^{L-1} \left(\tilde{w} c_{j+1,A}^\dagger c_{jB} + \text{h.c.} \right)$$

$$\tilde{v} = v \varphi(g, b_0)$$

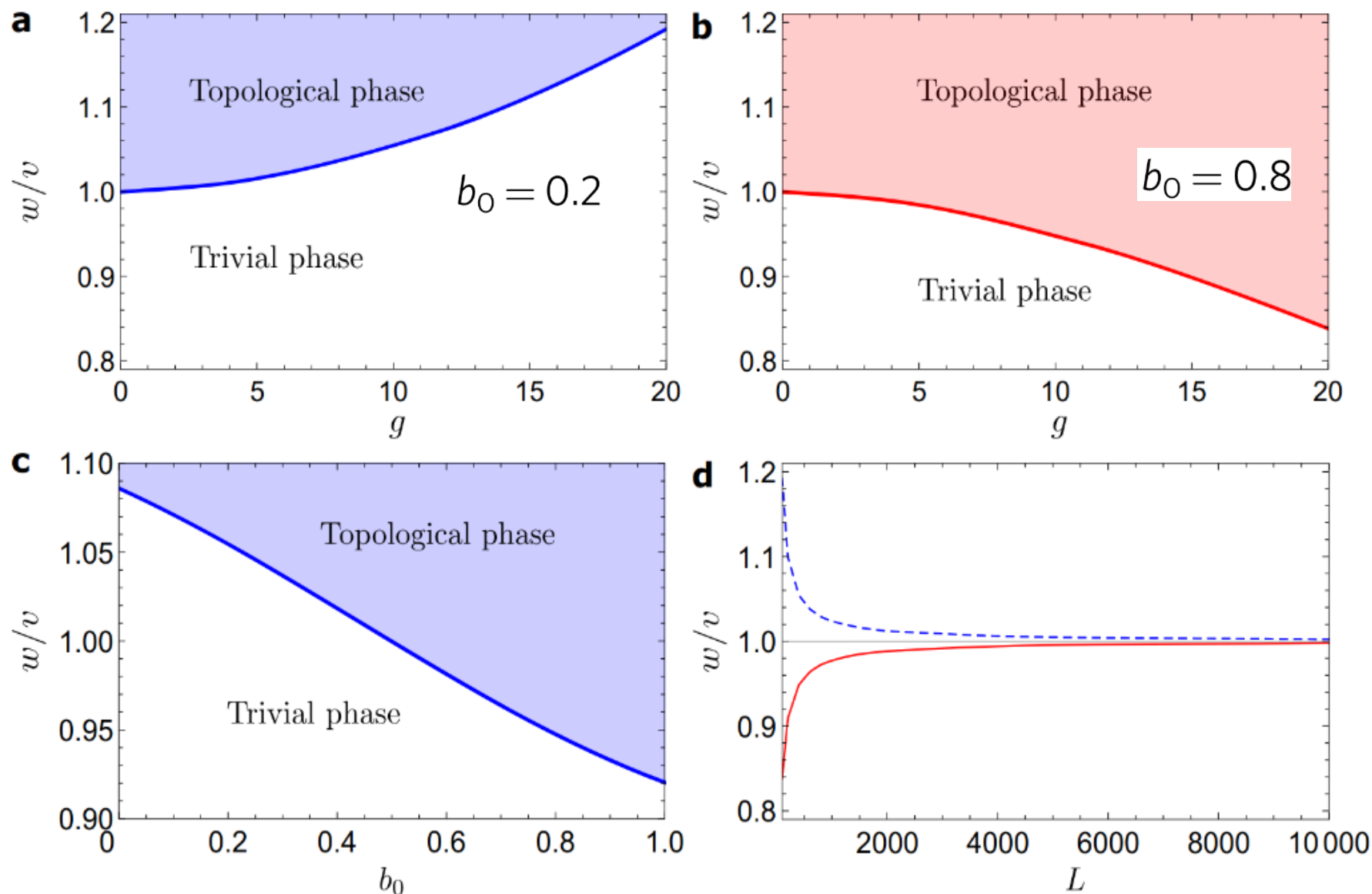
$$\tilde{w} = w \varphi(g, 1 - b_0)$$

$$\varphi(g, \ell) = \langle \phi | e^{-i \frac{g \ell}{\sqrt{L}} (a + a^\dagger)} | \phi \rangle$$

$$\tilde{w} / \tilde{v} = 1$$



Why a squeezed state?

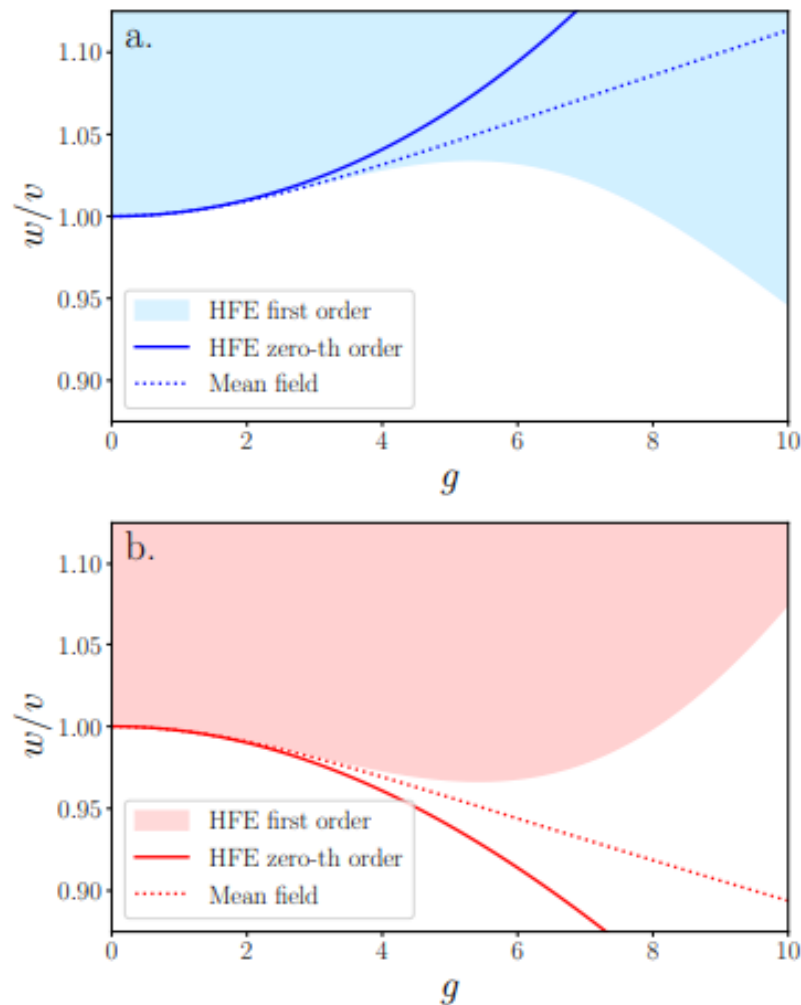


Coupling to the cavity can induce a transition

Floquet Theory of lattice electrons coupled to an off-resonant cavity

Jules Sueiro, Gian Marcello Andolina, Marco Schirò

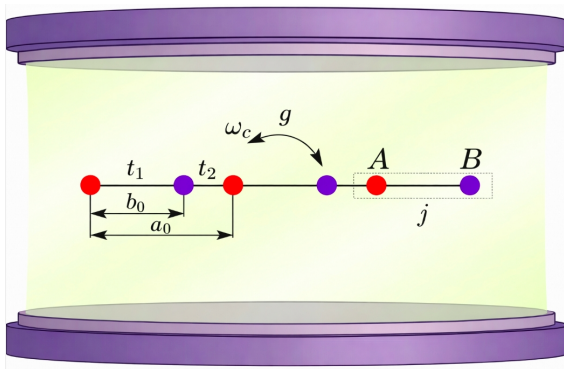
arXiv:2507.22715



The resonant case

$$H = \tilde{t}_1 \sum_{j=1}^L \left(c_{jA}^\dagger c_{jB} + \text{h.c.} \right) + \tilde{t}_2 \sum_{j=1}^{L-1} \left(c_{j+1,A}^\dagger c_{jB} + \text{h.c.} \right) + \omega_c \left(a^\dagger a + \frac{1}{2} \right)$$

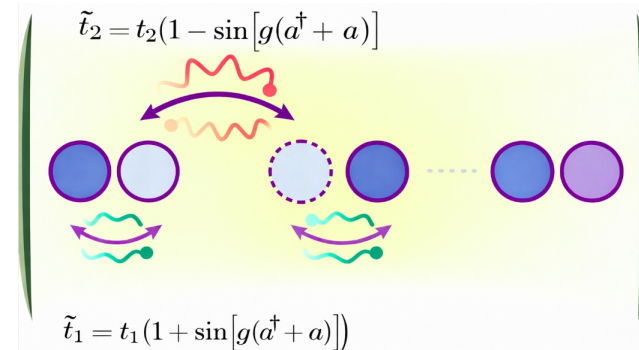
Peierls Substitution Coupling:



$$\tilde{t}_1 = t_1 e^{i \frac{g}{\sqrt{L}} b_0 (a + a^\dagger)}$$

$$\tilde{t}_2 = t_2 e^{-i \frac{g}{\sqrt{L}} (1 - b_0) (a + a^\dagger)}$$

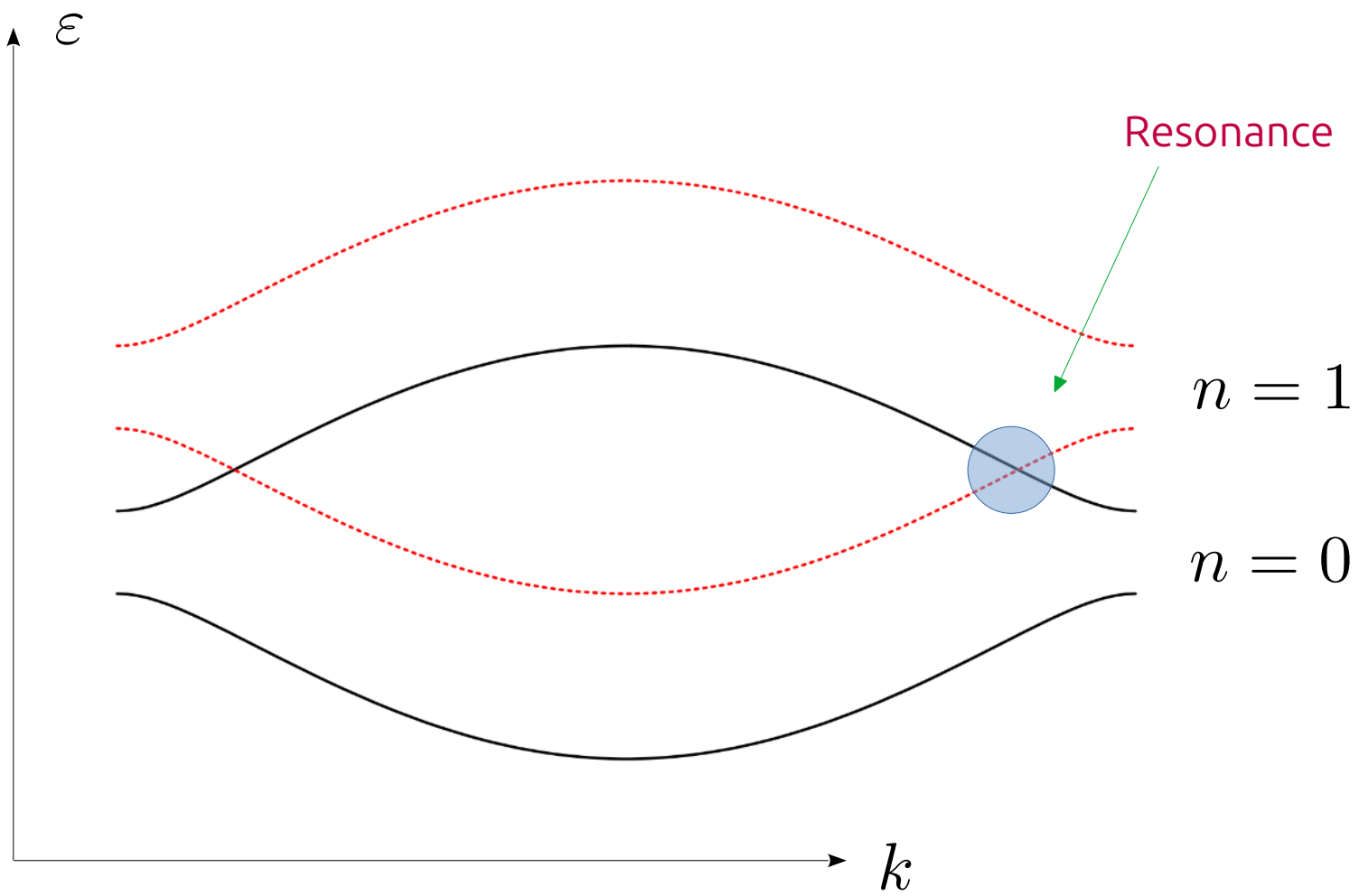
Phonon Modulation/transmon qubits



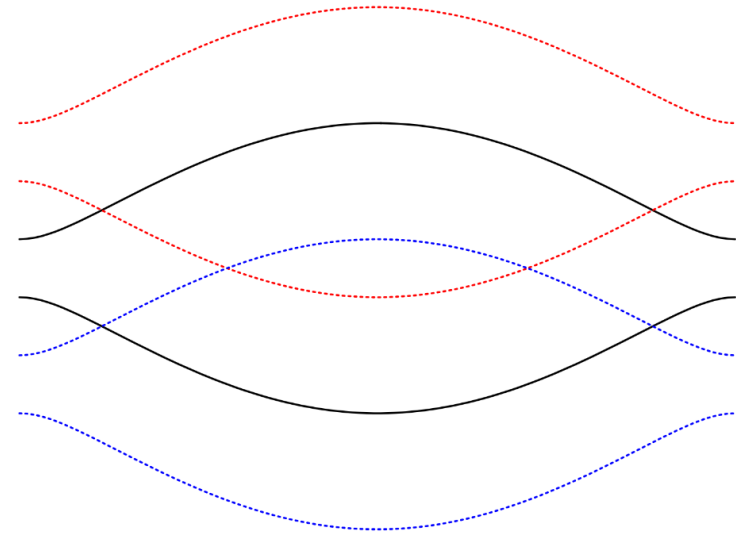
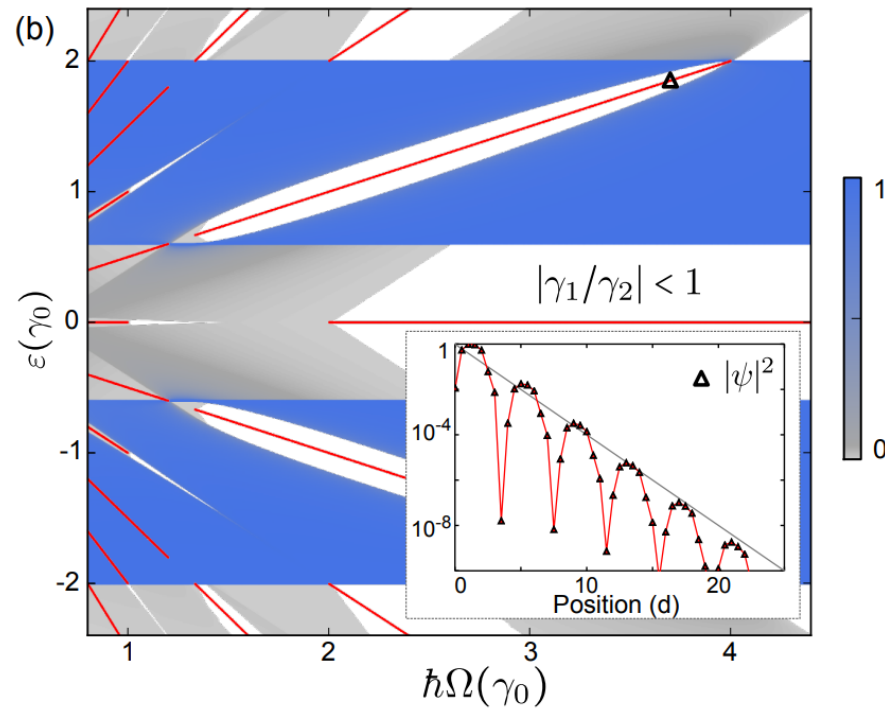
$$\tilde{t}_1 = t_1 (1 + \sin [g(a^\dagger + a)])$$

$$\tilde{t}_2 = t_2 (1 - \sin [g(a^\dagger + a)])$$

What kind of physics we look for?



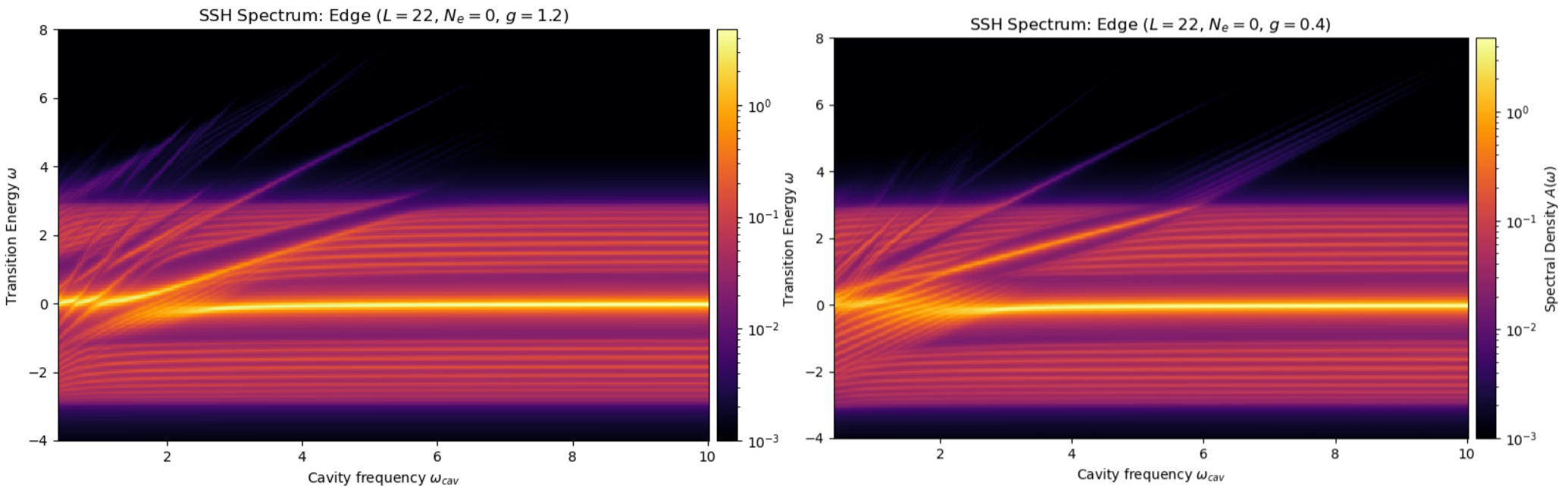
The Floquet case (for lattice distortion)



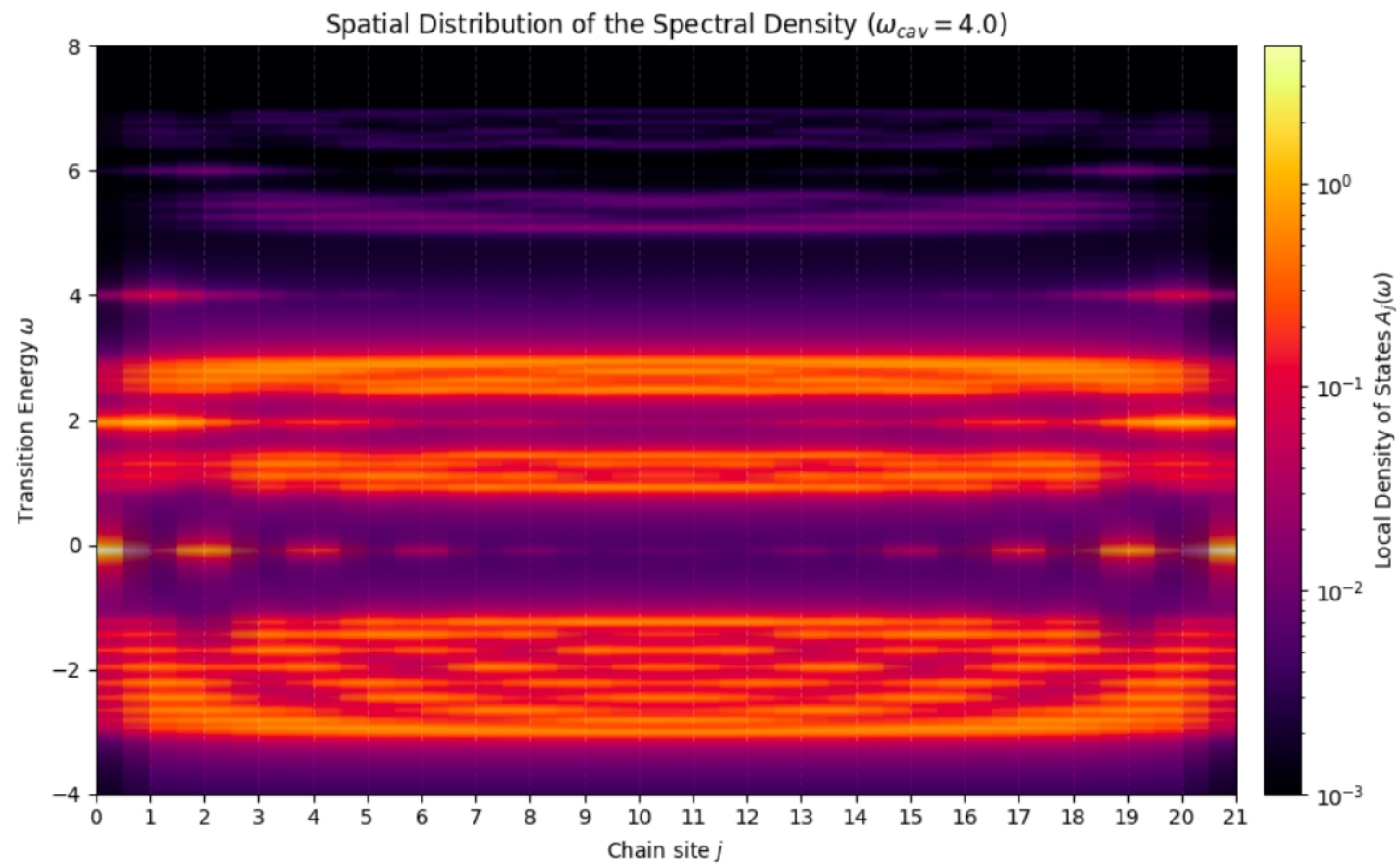
New edge states appear at $\varepsilon = \pm\hbar\Omega/2$

V. Dal Lago, M. Atala, and L. E. F. Foa Torres
Phys. Rev. A 92, 023624 (2015)

Back to the cavity case

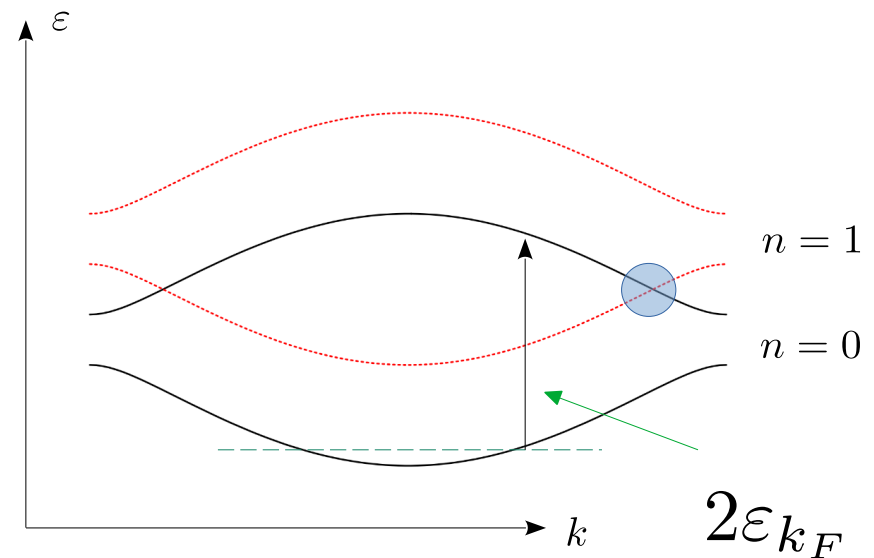
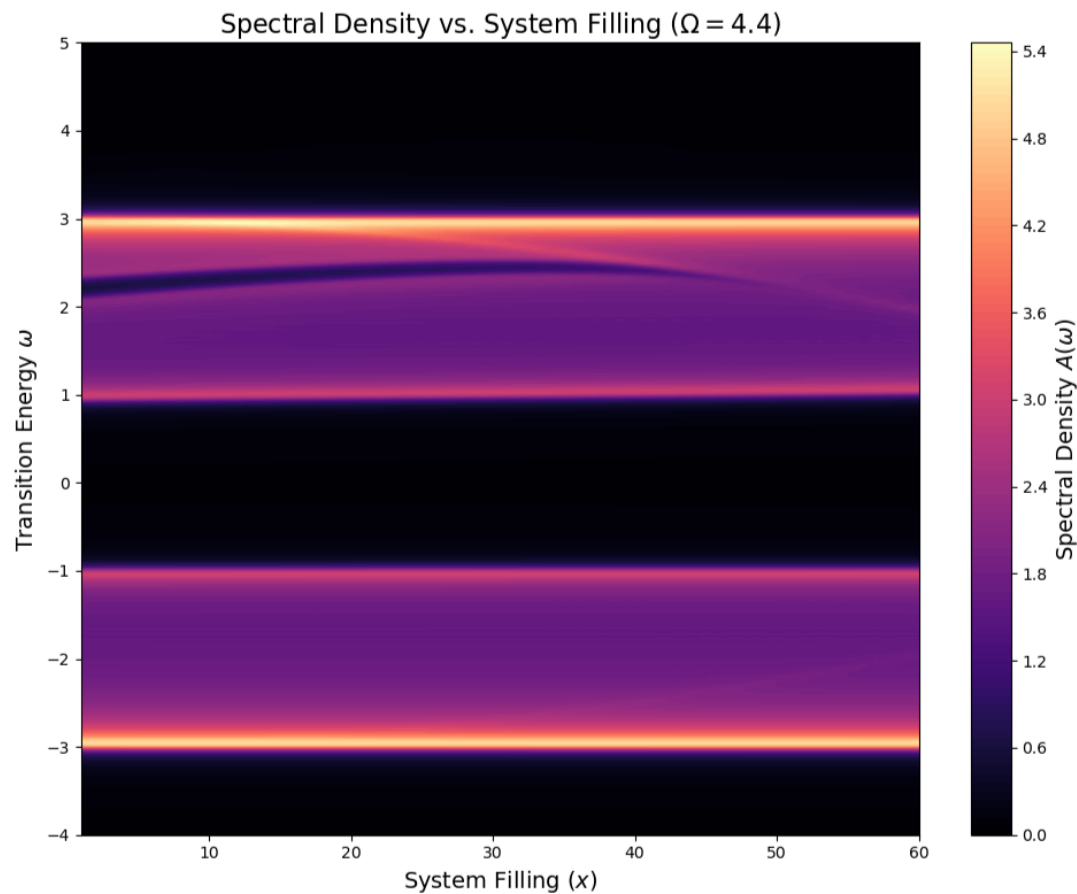


- Peierls $\rightarrow$ there is gap but not edge states
- Lattice distortion $\rightarrow$ there is a new edge states, induced only by the cavity!
- The edge states is a polariton induced by vacuum fluctuations!



The effect of the Fermi sea

- So far all work addressing many body systems work in the off-resonance situation
- Those that consider the resonant case work only with a single electron



Gap closing at $\hbar\Omega = 2\epsilon_{k_F} \rightarrow$ edge state is lost

V. Barroso, et al., *in preparation* (2026)

... and that is all...

Thanks